FH AACHEN
UNIVERSITY OF APPLIED SCIENCES

DEVELOPMENT OF A PROCEDURE FOR THE AUTOMATED PROVISION OF DESIGN-ACCOMPANYING MANUFACTURING INFORMATION TO ENHANCE MACHINING PRODUCTION EFFICIENCY

Bachelor Thesis

Submitted by

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Declaration of Authorship

I, Lumnwi Binla Clotilda Che, hereby declare that this thesis work has been conducted entirely on my own accord.

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Table of Contents

ACKNOWLEDGEMENTS ..... **I**

DECLARATION OF AUTHORSHIP ..... **II**

TABLE OF CONTENTS ..... **III**

ABSTRACT ..... **V**

ZUSAMMENFASSUNG ..... **VI**

1 INTRODUCTION **1.1**

1.1 General Introduction to the Topic 1.1

1.2 Aim and Structure of the Project 1.2

1.3 Brief Introduction of FVP and EXAPT Systemtechnik GmbH 1.3

2 LITERATURE REVIEW **2.4**

2.1 Computer-Aided Design (CAD) Systems 2.4

2.1.1 Definition *2.4*

2.1.2 How does Computer-aided Design work? *2.4*

2.1.3 Types of Computer-aided Design *2.5*

2.2 Computer-Aided Manufacturing (CAM) 2.6

2.2.1 Definition *2.6*

2.2.2 Machining Processes Automated by CAM. *2.6*

2.2.3 How CAM Systems work – From CAD to CAM *2.7*

3 STATE-OF-THE-ART COMMUNICATION BETWEEN A DESIGNER AND MANUFACTURER **3.8**

3.1 Communication between Engineering Designers and Manufacturers 3.8

3.2 Design for Manufacturing (DFM) 3.10

3.2.1 Definition of DFM *3.10*

3.2.2 The Five Principles of DFM *3.10*

3.2.3 Design for Manufacturing (DFM) Guidelines *3.11*

4 DEVELOPMENT OF THE EXPERIMENTAL PROCEDURE **4.13**

4.1 Overview 4.13

4.2 Computer- Aided Design of the Model 4.13

4.2.1 Design Parameters *4.14*

4.3 CAM Programming of the Wall Flange and Creation of Part Variations 4.15

4.3.1 Tools Employed in the Machining Process *4.16*

4.3.2 Machine Operations Performed to Produce the Final Part *4.16*

4.4 Derivation of the Key Performance Indicators for the Analysis of the Machining Processes 4.18

5 DATA ACQUISITION AND RESULTS 5.19

5.1 VARIATION IN THE NUMBER OF HOLES 5.20

5.2 NUMBER OF POCKETS 5.22

5.3 LENGTH OF THE SHAFT 5.24

6 EVALUATION OF THE PART AND GUIDELINE PROPOSAL 6.27

6.1 EVALUATION OF THE CAD PARAMETERS AND PROPOSAL OF GUIDELINES 6.27

6.2 IMPLEMENTATION OF THE GUIDELINES IN THE CAD MODEL 6.28

6.3 IMPLEMENTATION OF GUIDELINES ON THE MACHINING OPERATIONS 6.29

6.4 DATA ACQUISITION AND RESULTS AFTER IMPLEMENTATION OF THE PROPOSED GUIDELINES 6.30

6.4.1 Number of Holes 6.30

6.4.2 Number of Pockets 6.33

6.4.3 Length of Shaft 6.37

7 PROCEDURE PROPOSAL 7.45

7.1 OVERVIEW 7.45

7.2 SYSTEM FRAMEWORK 7.46

7.3 FEEDBACK CONTENT AND ACQUISITION STRATEGY 7.47

8 CONCLUSION 8.48

8.1 OVERVIEW 8.48

8.2 LIMITATIONS 8.48

8.3 SUGGESTIONS FOR FUTURE WORK 8.49

9 BIBLIOGRAPHY 9.50

LIST OF FIGURES VII

LIST OF TABLES VIII

Abstract

The manufacturing industry has undergone significant changes in recent years, driven by increased competition and the need to improve efficiency and reduce costs. In this context, the design process plays a critical role in setting the foundation for the final product. As a result, there has been growing emphasis on developing methods that can streamline and optimize the design process, with the aim of improving efficiency and reducing the overall cost of manufacturing.

One key area of focus in this regard is the provision of manufacturing information during the design process. Manufacturing information provides details on the specifications of the component being designed, including its geometry, dimensions, material properties, and manufacturing requirements. This information is crucial for the production process, as it enables designers to create designs that are optimized for production and reduces the risk of errors or inefficiencies in the manufacturing process. However, the process of providing manufacturing information has traditionally been a time-consuming and manual process, which can lead to delays and errors in the manufacturing process.

To address this challenge, this work aims to develop a method for the automated and design-accompanying provision of manufacturing information to the designer. The proposed method will be derived and evaluated through a case study, where specific indicators will be analysed and compared to results obtained from traditional design methods. This comparative analysis will aid in coming up with the procedure and subsequent software design and will involve a detailed examination of the efficiency, accuracy, and cost-effectiveness of the proposed method.

This proposed procedure when developed may be designed to integrate existing Computer-aided Design (CAD) software, enabling designers to receive manufacturing information in real-time as they design components, reducing the effort required for the machining production of the component they have designed. By providing this information in an automated and design-accompanying way, the production process will be streamlined improving efficiency.

This work will provide a detailed exploration of the proposed method, including its design, development, and evaluation. The potential implications and contributions of the proposed method for the manufacturing industry will also be discussed.

Ultimately, this work aims to contribute to the ongoing efforts to streamline and optimize the design process in the manufacturing industry.

Zusammenfassung

Die Fertigungsindustrie hat sich in den letzten Jahren stark gewandelt, bedingt durch den zunehmenden Wettbewerb und die Notwendigkeit, Effizienz zu steigern und Kosten zu senken. In diesem Zusammenhang spielt der Designprozess eine entscheidende Rolle bei der Gestaltung der Rahmenbedingungen für das Endprodukt. Infolgedessen wird immer mehr Wert auf die Entwicklung von Methoden gelegt, mit denen der Designprozess rationalisiert und optimiert werden kann, um die Effizienz zu steigern und die Gesamtkosten der Herstellung zu senken.

Ein wichtiger Schwerpunkt in diesem Zusammenhang ist die Bereitstellung von Fertigungsinformationen während des Designprozesses. Fertigungsinformationen liefern Details zu den Spezifikationen des zu entwerfenden Bauteils, einschließlich seiner Geometrie, Abmessungen, Materialeigenschaften und Fertigungsanforderungen. Diese Informationen sind für den Produktionsprozess von entscheidender Bedeutung, da sie es den Konstrukteuren ermöglichen, Designs zu erstellen, die für die Fertigung optimiert sind, und das Risiko von Fehlern oder Ineffizienzen im Fertigungsprozess verringern. Die Bereitstellung von Fertigungsinformationen ist jedoch traditionell ein zeitaufwändiger und manueller Prozess, der zu Verzögerungen und Fehlern im Fertigungsprozess führen kann.

Um dieser Herausforderung zu begegnen, zielt diese Arbeit darauf ab, eine Methode für die automatisierte und konstruktionsbegleitende Bereitstellung von Fertigungsinformationen für den Konstrukteur zu entwickeln. Diese vorgeschlagene Methode wird anhand einer Case Study und der Analyse spezifischer Kennzahlen abgeleitet und bewertet, deren Ergebnisse, mit denen der traditionellen Konstruktionsmethoden verglichen werden. Die vergleichende Analyse wird eine detaillierte Untersuchung der Effizienz, der Genauigkeit und der Kostenwirksamkeit der vorgeschlagenen Methode beinhalten.

Das vorgeschlagene Verfahren wird so konzipiert sein, dass es bestehende CAD-Software integriert und es den Konstrukteuren ermöglicht, Fertigungsinformationen in Echtzeit zu erhalten, während die Bauteile konstruiert werden, wodurch der Aufwand für die Fertigung des konstruierten Bauteils verringert wird. Durch die automatisierte Bereitstellung dieser Informationen wird der Produktionsprozess rationalisiert und die Effizienz gesteigert.

In dieser Arbeit wird die vorgeschlagene Methode, einschließlich ihrer Konzeption, Entwicklung und Bewertung, eingehend untersucht. Darüber hinaus werden die potenziellen Auswirkungen und Beiträge der vorgeschlagenen Methode für die Fertigungsindustrie erörtert. Letztlich soll diese Arbeit einen Beitrag zu den laufenden Bemühungen um die Rationalisierung und Optimierung des Konstruktionsprozesses in der Fertigungsindustrie leisten.

1 Introduction

1.1 General Introduction to the Topic

In modern manufacturing, achieving maximum efficiency and productivity while maintaining the highest quality standards is essential. To achieve this, manufacturers must ensure that every stage of the manufacturing process is optimized, from design to production. One of the most critical stages in this process is the provision of manufacturing information to the designer during the design phase. Traditionally, this process has been manual and time-consuming, leading to delays in production and increased costs. However, recent advances in computer-aided design (CAD) and computer-aided manufacturing (CAM) technologies have opened up new possibilities for the automated provision of manufacturing information.

This project aims to develop a procedure for the automated provision of design-accompanying manufacturing information to enhance machining production efficiency. By automating this process, the designer can receive the necessary manufacturing information in real-time, reducing the time and effort required for machining production. This work will investigate the use of CAD and CAM technologies in developing this procedure, as well as the analysis of key performance indicators (KPIs) to ensure that the process is optimized for maximum efficiency.

Finally, this project aims to contribute to the development of a more efficient and streamlined manufacturing process, reducing costs, improving quality, and increasing productivity.

1.2 Aim and Structure of the Project

The objective of this project is to develop a comprehensive method for the automated provision of manufacturing information to the design engineer during the design process. The method will assist the design engineer in reducing the effort required for the machining production of the component designed. The proposed method aims to address a significant challenge in the manufacturing industry: the disconnect between the design and manufacturing phases.

The overall project will involve several phases. The first phase, on which this work is based, will include the development of the proposed method which will incorporate the collection of data and the identification of the critical manufacturing information required by designers during the design process.

The work comprises in following tasks:

- The creation of sample components from a selected range of components using SolidWorks under consideration of CAD software parameters
- The automated CAM machining planning in EXAPT-Solid using feature recognition and toolpath planning algorithms.
- Derivation of Key Performance Indicators (KPIs) to evaluate part manufacturability.
- The conceptual design and implementation of a CAM-integrated result representation
- Development of a procedure for deriving spectrum-specific component design guidelines for the design

The second phase may involve the development of the software, including the design of the user interface, based on the KPIs analysed and the developed proposed procedure from the first phase. The software may be designed to provide real-time feedback on manufacturing feasibility and tool path optimization, ensuring that the design engineer has all the necessary data required for efficient machining production. To ensure that the software is user-friendly and intuitive, extensive research may be conducted on the needs of the design and manufacturing engineers. The software may be designed to be compatible with existing CAD software, allowing for easy integration and reducing the learning curve.

1.3 Brief Introduction of FVP and EXAPT Systemtechnik GmbH

The research association Forschungsvereinigung Programmiersprachen für Fertigungseinrichtungen (FVP) is an association committed to the idea of carrying out application-oriented research and the rapid transfer of research results into industrial manufacturing applications.

The main objective of its efforts is to develop viable and sustainable solutions that are adaptable to the evolving needs of production facilities, structures, and user interactions, with a special emphasis on maximizing the benefits of CAD/CAM systems within an effective communication network.

FVP is in close cooperation with EXAPT Systemtechnik GmbH, the managing enterprise of the EXAPT (EXtended Subset of Automatically Programmable Tools) programming language, which is a programming language used to generate Numerical Control (NC) programs with control information for machining tools and enables to consider production-related issues of various machining processes.

EXAPT offers versatile software solutions for the manufacturing industry, with two primary product groups: EXAPTcam and EXAPTpdo. These products can be incrementally expanded to accommodate individual needs and are compatible with existing Information Technology (IT) infrastructures.

EXAPTcam is specifically designed for NC planning, supporting operations like turning, drilling, and milling up to 5-axis simultaneous machining. It can import data from various sources, including 3D CAD models, drawings, and tables. Additionally, it generates manufacturing information such as clamping and tool plans, pre-setting data, and time calculations. The software's realistic simulation capabilities ensure reliable production.

EXAPTpdo, or EXAPT ProductionsDataOrganisation, offers a technology platform that is compatible with various systems within the value chain. It synchronizes NC planning and shop floor systems and enables faster communication, facilitating greater flexibility in production.

2 Literature Review

2.1 Computer-Aided Design (CAD) Systems

2.1.1 Definition

Computer-aided Design (CAD) refers to the utilization of computers or workstations to generate designs and simulations of real-world products, which can then be digitally modified, analysed, and perfected in terms of scale, precision, and physical properties. (Wikipedia, 2023) CAD software aids in the increase of designer productivity, improvement in communication through documentation, and the creation of a manufacturing database. CAD data is often exported in the form of electronic files for printing, machining, or other production operations.

CAD software in Mechanical Engineering involves more than just the design of the tool. It involves the use of either vector-based graphics to depict the objects of manual drafting or the production of raster graphics showing the overall view of the designed objects. The output of CAD must, similarly to manual drafting or technical drawing, contain information, useful for the machining of the product, such as geometry, dimensions, materials, processes, and tolerances, according to application-specific conventions.

2.1.2 How does Computer-aided Design work?

The first step in the CAD process involves conceptualizing the design of the object or structure to be created. This can involve sketching rough ideas or blueprints of the design, which can then be imported into the CAD software. Once the concept has been developed, the next step is to create a digital model of the object or structure using the CAD software. This is typically done by creating a 2D or 3D representation of the design, which can then be refined and modified using a variety of tools and features.

After creation of the model, it can be analysed using specialized algorithms and modelling techniques to simulate real-world physics, such as gravity, friction, and material properties. This analysis can help designers and engineers identify potential problems or areas for improvement in the design.

The science behind CAD is based on a variety of disciplines, including mathematics, physics, and computer science. CAD software uses mathematical algorithms to create precise and accurate digital models of physical objects or structures. It also incorporates physics-based simulations to predict how the design will behave in the real world, allowing designers and engineers to optimize the design for performance, durability, and other factors.

2.1.3 Types of Computer-aided Design

There are different types of CAD, each needing the operator to think differently about how they are used and how differently their virtual components are designed.

- **2D CAD**

This is the earliest iteration of CAD, created in the early 1970s. It mainly uses text, dimensions, leaders, and tables to annotate technical drawings. 2D CAD is used to design, plan, section, and detail structures in the built environment. Furthermore, 2D CAD shows how various components work together to create assemblies and offer essential inspection locations. AutoCAD, CADkey, and CATIA V4 are some examples of 2D CAD applications.

- **3D CAD**

3D CAD produces a realistic representation of the design object. This type of CAD enables designers to view and work on potential issues earlier in the production chain with lower production costs. Examples of 3D CAD software include Inventor by Autodesk, CATIA V5, SolidWorks, and VX CAD.

3D CAD is further divided into 3 categories.

1. **Surface Modelling**: This is the most advanced and accurate type of CAD. When it comes to the complexity of a part, surface modelling goes a lot further. It uses guidelines to specify the outline and the curvature of the design object. It is most used for illustrations and architectural representations since it can readily integrate all necessary features. Visual representations that are impossible to obtain in the real world can be provided using surface modelling. However, it is crucial to establish whether such a design can be manufactured physically. CATIA, Inventor, and Free CAD are examples of surface modelling CAD software.
2. **Wireframe Modelling:** Here, shapes are represented as a network of vertices. At least three vertices, each of which has one appearance, make up a geometric face. As each vertex's position changes, so do the corresponding item's sizes and statuses. Triangles are a common building block in wireframes for devices, and the more triangles there are, the more realistic the wireframe appears to be. Autodesk Maya and Cinema 4D are examples of software that use Wireframe Modelling.
3. **Solid Modelling**: This type of CAD is useful when dealing with flat surfaces or simple bends with regular radii. Also, it's conducive to precise measurements and angles. It entails basic shapes such as circles, cubes, and n-sided prisms. Some designers start with two-dimensional sketching and then transform the sketch into a three-dimensional figure, while others build complicated figures by layering solids on top of solids. SolidWorks and Autodesk Fusion are good examples of this.

2.2 Computer-Aided Manufacturing (CAM)

2.2.1 Definition

Computer-Aided Manufacturing (CAM), also known as computer-aided machining, is the application of software for controlling machine tools in the production of workpieces. Alternatively, it can refer to the utilization of computer technology to aid in all aspects of a manufacturing facility's operations, such as planning, management, transportation, and storage. Its main goal is to speed up manufacturing and develop parts and tools with more accurate dimensions and material consistency, using less energy, while in some situations, using only the necessary amount of raw material (thus minimizing waste).

CAM is a successive process after CAD, as the part created in CAD (and sometimes verified in CAE – computer-aided Engineering) is imported into CAM software, which converts it into a machine language that the machining object can understand, and then proceeds with the production.

2.2.2 Machining Processes Automated by CAM.

Machining in CAM occurs with the help of CNC (Computer Numerical Control) machines. These machines manufacture a part by using G and M codes provided by the CAM software. The following processes can be automated by CAM.

- **Milling:** Here excess raw material is precisely removed to reveal the final workpiece. This is an essential form of subtractive manufacturing.
- **Turning:** The workpiece is rotated against a machining tool to remove excess raw material. Turning is another form of subtractive manufacturing.
- **Electrical discharge machines:** Here parts are created by firing an electric spark through them. These sparks may readily cut through any material because of the extraordinarily high temperatures they can attain. With the help of CAM, these sparks can be finely regulated to precisely cut the workpiece.
- **CNC routers:** They work similarly to milling machines, removing excess raw material to reveal the final workpiece.
- **3D-Printing:** With 3D Printing, CAM can produce a workpiece virtually by adding successive layers of compatible materials until the desired shape is complete.
- **Water, Plasma, and Laser Cutting:** This process results in a fine cut or engraved finish using accurate lasers, high-pressure water, or a plasma flame. These machines can finish the same work in hours or days that manual engraving processes take months to complete. Plasma cutters are essential for cutting through electrically conductive materials like metals.

2.2.3 How CAM Systems work – From CAD to CAM

For a model to be produced, it must first be conceptualized and planned. This is part of the product development life cycle before the actual production of the part. An accurate draft of the plan must be drawn before actual production takes place.

The first stage of the production process is the design stage. In this phase, the engineering designer uses CAD software to produce the part, either in 2D or 3D. The part's functionality, manufacturability, and aesthetics are the main considerations here.

After the creation of the model, the designer assigns coordinates to the part, enabling the use of the CAD software's transformation features. Subsequently, a production simulation is performed, to evaluate the part's feasibility considering the setup's manufacturing capabilities.

Computer-aided manufacturing is the next step after the design is finished. The CAD software exports the finished model, together with the design data to the CAM software. If the software has both CAD and CAM capabilities the export and import of the drawings are not required.

After the import is complete, the CAM software starts preparing the part for the manufacturing process. It starts by looking for mistakes in the model and subsequently creates a CNC program to produce the physical part. The CNC program is essentially a set of coordinates that directs the cutting head of a tool, per machining object, during the manufacturing process. Selecting the appropriate settings for each machining object is the next stage. These settings include cutting speed, voltage, feed, and revolutions per minute (RPMs). It is also important to note that the geometry of the part as well as the equipment and tooling available, all affect the model.

Finally, the software determines the nesting. Nesting refers to how components are positioned, and oriented about raw materials. The purpose is to maximize material usage.

When all of this is input into the CAM software, it translates the information into codes that the machine can comprehend. These include the Geometric code (G-code) and the Machine/Miscellaneous code(M-code).

The G-code determines the movement and speed of the cutting heads, while the M-code provides supplementary information overlooked by the G-code. Once the codes are created, they are loaded into the machine, which automatically does the job of executing the codes to transform a raw material block into a finished product.

3 State-of-the-Art Communication Between a Designer and Manufacturer

3.1 Communication between Engineering Designers and Manufacturers

To help designers in their decision-making for the acquisition of optimal CAD designs, manufacturers might have to offer information, advice, or feedback, including DFM guidelines. Thus, it is important to study how designers and manufacturers communicate.

Drawings and CAD models are the primary modes of communication between a designer and a manufacturer. Despite the inclusion of additional layers to the drawing, such as notations on the drawing's geometry, dimensioning, and tolerance, as well as other informal records, models transferred between designers and manufacturers may be misconstrued because of the viewers' varying points of view.

In addition to CAD drawings and notations, verbal communication and gestures are also used, yet because of the increasingly global nature of the manufacturing industry, these extra modes of communication may be expensive or challenging. Also, inaccuracies or miscommunication during the design and manufacturing stages of development might be brought on by differences in interpretation of the gestures and words of one of the parties.

One remedy to this problem was real-time collaboration between the designers and manufacturers so that they could jointly contribute to the shape design while making it ideal for manufacturing. (Song, 2019) Early collaboration between manufacturers and designers can drastically lower costs and raise quality. However, before they can work together, they must find a common ground. The CAD model could be treated as a boundary object; an object with constant properties that may be perceived or used differently by different participants depending on their expertise or goals, and the interpretations of the CAD model would have to be compatible (or similar, not identical) for effective design communication to take place. However, there are chances of incompatibility in interpretation where the designer’s interpretation doesn’t match the manufacturer’s, resulting in inconsistent design communication and hence prolonging the time needed for the manufacturing of the design.

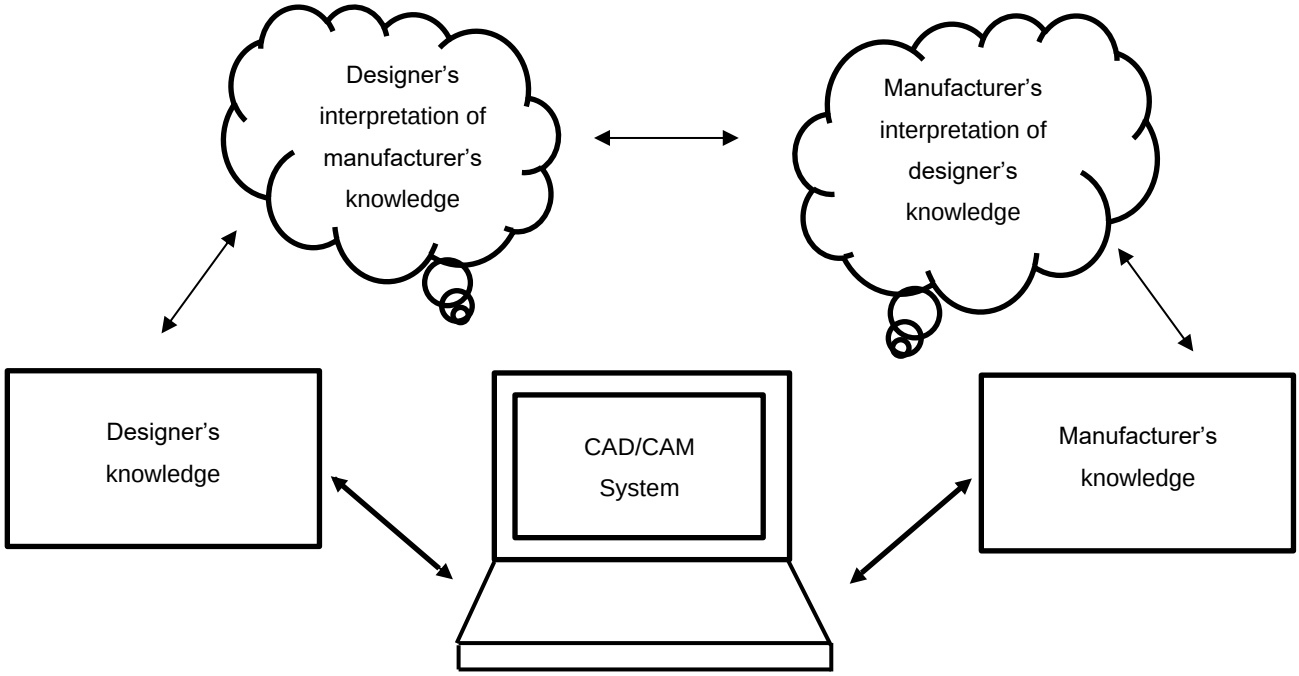


Figure 1: Communication Between Designers and Manufacturers (Song, 2019)

3.2 Design for Manufacturing (DFM)

3.2.1 Definition of DFM

Design for Manufacturing (DFM) refers to the approach of designing parts, components, or products with the aim of simplifying, optimizing, and refining the product design for ease of manufacturing, ultimately resulting in a better product at a reduced cost of time and fabrication. DFM is a design philosophy that guides designers toward creating good designs from a manufacturing standpoint. DFM breaks the isolation that exists between design and manufacturing by taking manufacturing constraints into account at every level of the design process. Conceptually, DFM is an alternative solution to problems that can arise later in the manufacturing process and is carried out sooner in the design stage.

3.2.2 The Five Principles of DFM

Five principles are examined during DFM. They include:

1. **Process:** It's crucial to use the proper manufacturing process. To manufacture goods at a reasonable price, the process should employ lower-capitalized techniques. It ought to be adjusted to each component's dimensions. Tolerances and the required materials should be considered during the process.
2. **Design:** The ideal design is typically one that is straightforward. In terms of price, production, application, and maintenance. Nonetheless, good manufacturing practices should be followed in all designs. The design must include more details, such as thicknesses, tighter tolerance requirements, texture, and consistency.
3. **Materials:** Time and money can be saved by making the appropriate material selections early on. Both price and quality are impacted by the materials chosen. While picking materials, a few factors come into play. Mechanical qualities, flammability, conductivity, and thermal properties are a few to consider.
4. **Environment:** Each part of the product must be constructed with consideration for the environment in which it will be used. Every operational condition must be considered. Each component of the product must function properly under these circumstances.
5. **Testing:** Finally, testing needs to be detailed. At every level, all products and components must adhere to industry standards. These standards may be organizational, internal, or industry specific. All DFM stages must take these standards into account.

3.2.3 Design for Manufacturing (DFM) Guidelines[1]

A set of design principles or rules organized to assist the designer in lowering the cost and difficulty of manufacturing an item forms the basis of any design for the manufacturing system. These regulations are listed below.

1. **Reduce the total number of parts:** Fewer parts mean less need for purchases, inventory, handling, processing, equipment, engineering time, difficult assembly, service inspection, or testing. In general, it lessens the level of intensity of all activities involving the product over its whole life. A portion that doesn't need to move relative to other parts, doesn't need to be manufactured of a different material or would make assembling or maintaining other parts very challenging or impossible is a great candidate for deletion. One-piece constructions and the use of manufacturing processes including injection moulding, extrusion, precision casting, and powder metallurgy are some strategies for part-count reduction.
2. **Develop a modular design:** Modular product design makes it easier to perform tasks like inspection, testing, assembly, purchase, redesign, maintenance, and service. One explanation is that modules increase product update adaptability during the redesign phase, aid in testing before final assembly, and enable the use of standard components to reduce product variation.
3. **Use of standard components:** Standard components are cheaper than custom-made goods. Product lead times are shortened by the high availability of these components. Their reliability characteristics have also been thoroughly established. Also, the use of standard components shifts the burden of production on the supplier, partially easing the manufacturer's worry about keeping to production deadlines.
4. **Design parts to be multi-functional:** By reducing the overall number of pieces in a design, multi-functional parts achieve the advantages mentioned in rule 1. Examples include a part that serves as both a structural member and an electric conductor or one that serves as both a heat dissipation element and a structural member. Apart from their primary use, some pieces may also have guided, aligning, or self-fixturing characteristics to make assembly easier, and/or reflective surfaces to make inspection easier.
5. **Design parts for multi-use:** In manufacturing, parts that are intended for several uses can be shared by other products. When utilized in various products, these components may serve the same or distinct purposes.
6. **Design for easy fabrication:** To reduce the overall cost of manufacturing, choose the material and fabrication method that work well together. Final processes like painting, polishing, and finishing machining, should generally be avoided. Problems like excessive tolerance, surface-finish requirements, and others are frequently encountered and lead to production costs that are higher than necessary.

7. Avoid separate fasteners: Due to the handling and feeding activities that must be carried out, using fasteners raises the cost of manufacturing a part. Fasteners should generally be avoided and substituted with other design elements, like tabs or snap fits. If fasteners must be used, there are some guidelines that should be observed while choosing them. Reduce the number, size, and variety utilized as little as possible. If possible, use standard components.

8. Minimize assembly directions: It is best to assemble all parts in one direction. The optimum way to add parts is from above, vertically, and in a direction that is opposite to the direction of gravity, if at all possible (downward). By choosing this direction, gravity's effects assist the assembly process rather than needing to be compensated for when other directions are used.

9. Maximize compliance: Due to differences in part size or the precision of the positioning device being employed, errors can happen during insertion procedures. The equipment and/or the part may be damaged as a result of this defective behaviour. Due to this, compliance must be taken into account when designing parts and assembling them. Compliance in the assembly process can be seen in the choice of a rigid-base part, tactile sensing abilities, and vision systems. Using high-quality parts with designed-in compliance, a rigid base part and selective compliance in the assembly tool is a straightforward approach.

10. Minimize Handling: Positioning, orienting, and fastening a part or component are all parts of handling. When possible, symmetrical elements should be employed to aid with orientation. If that cannot be done, then the asymmetry must be made more extreme to prevent failures. Employ external guiding features to assist with part orientation. The subsequent operations must be planned to keep the part alignment. Try to reduce the flow of materials, parts, and other waste during the product design process. Also, consider packing when doing so, and use appropriate and secure packaging for the product.

[1]Design for Manufacturing – Guidelines

4 Development of the Experimental Procedure

4.1 Overview

This work involves four distinct stages, each with its specific objectives and tasks. The first stage involves the use of computer-aided design (CAD) software to create the design of the part. The second stage involves the CAM programming of the part, where the manufacturing process is planned out, and the necessary toolpaths are generated. The third stage involves the analysis of the manufacturing process, with the consideration of key performance indicators (KPIs) such as manufacturing time and number of tool changes. In this stage, the process is optimized to achieve maximum efficiency while maintaining quality. The final stage of the project involves the development of guidelines based on the findings from the previous stages, which can be used to inform future manufacturing processes.

4.2 Computer- Aided Design of the Model

The design of a part requires careful consideration of several factors such as aesthetics, functionality, and manufacturability. Aesthetics ensure that the part meets the desired appearance and visual appeal. Functionality ensures that the part is fit for its intended purpose while manufacturability ensures that the part can be produced economically and within the required time frame. The aim of this section is to design a part, a wall flange, that is aesthetically pleasing, functional, and easy to manufacture.

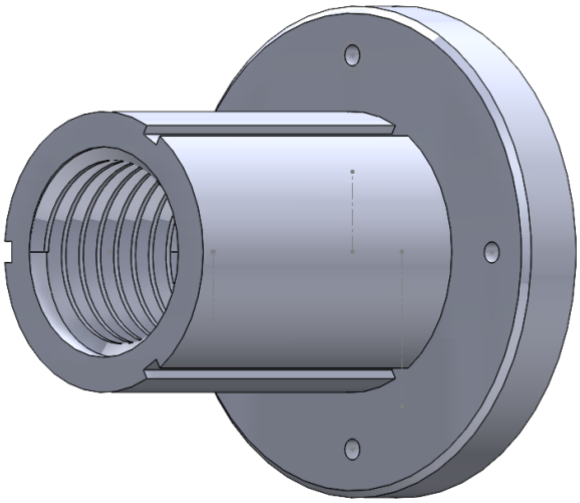


Figure 2: Initial Design of the Flange

The design of the wall flange considered three factors: aesthetics, functionality, and manufacturability. Aesthetics were given consideration to ensure that the flange looks appealing, visually attractive, and has a sleek design. The flange is a visible component, and its design can have a significant impact on the overall appearance of the product. Therefore, careful consideration was given to the design to ensure it fits the aesthetic requirements.

Functionality was also an essential consideration in the design of the flange. The flange is commonly used in piping systems, where it serves as a means of connecting pipes or tubes to each other, or to other equipment such as valves, pumps, or tanks. The flange needed to be designed with the appropriate dimensions to ensure that it can provide a secure and leak-proof connection between the two parts while allowing for easy assembly and disassembly when necessary.

Manufacturability was also a crucial consideration in the design of the wall flange. The design had to be optimized to ensure that it can be manufactured with ease and within the required time frame. The flange was designed to ensure that it could be machined using a CNC milling machine. This required careful consideration of the design parameters to ensure that the part could be manufactured economically.

4.2.1 Design Parameters

The design of the wall flange was carried out using SolidWorks, SolidWorks is a computer-aided design software that enables designers to create 3D models of parts and assemblies. Three design parameters were considered: the pockets on the shaft of the flange, the holes on the base of the flange, and the length of the flange's shaft.

The pockets on the shaft of the flange were added for aesthetic purposes. They were designed to add a decorative element to the flange and to make it look visually appealing. The size and shape of the pockets were carefully chosen (l = 1 mm) to ensure that they did not compromise the strength and functionality of the flange.

The holes on the base of the flange were designed to attach the flange to its connecting part. The diameter of the holes (ϕ2mm) was carefully scrutinised to ensure that they provided a secure attachment and did not compromise the strength of the flange. The spacing of the holes was also considered to ensure that the flange was balanced and did not cause any undue stress on the door.

The length of the flange's shaft was another design parameter that was analysed. The length of the shaft was carefully chosen to ensure that it provided sufficient support to the connecting part without being too long or too short.

4.3 CAM Programming of the Wall Flange and Creation of Part Variations

The process of CNC machining is a crucial step in the manufacturing of a part, and the CAM software plays an important role in this process. The part designed in SolidWorks is transferred to the CAM software, where a production plan is created. The CAM software used in this study is EXAPTsolid, which is a plug-in for SolidWorks. One of the advantages of having EXAPTsolid as a plug-in for SolidWorks is that there is no need for importing and exporting of the CAD model. This saves time and reduces the chance of errors that can occur during the import/export process.

To manufacture the flange, a drilling and milling machine was chosen (Machine name: EXABAZ, dc35) as the appropriate machine for the production process. This machine allowed for optimal machining operations to be performed from every side of the part. In addition, the raw material that was simulated for the production process was a 40mm x 60mm cylindrical cube, which was chosen to correspond to the geometry of the flange.

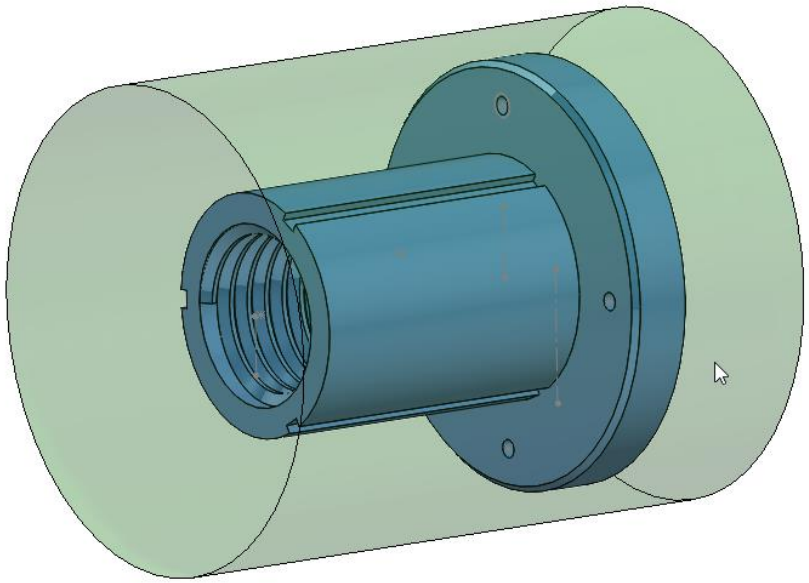


Figure 3: Simulation of the Raw Part

This allowed for accurate testing of the manufacturing process and ensured that the final product would meet the required specifications. The use of a cylindrical cube for the raw part was also beneficial in terms of reducing material waste and improving efficiency during the manufacturing process.

4.3.1 Tools Employed in the Machining Process

The final part was generated from the raw part using a set of tools with different functions. The table below summarizes the tools used in the process, along with their respective tool IDs according to the EXAPT database.

Tool ID	Tool Name	Function
6310160	End mill	Used for making shafts, shapes, and holes on the workpiece
63100100	End mill	Used for making shafts, shapes, and holes on the workpiece
631410	Twist drill	Used for side milling operations on the workpiece
63140140	Twist drill	Used for chamfering the edges of the workpiece
10008	Tapping Tool (metric normal)	Used for creating internal threads in the workpiece
6356020	Full Radius End Mill	Used for making holes in the workpiece

Table 1: Tools and Corresponding IDs in EXAPT Database

It is important to note that each tool was selected based on its intended function and suitability for the specific machining operation required for the part. The combination of these tools and their efficient use in the machining process resulted in the successful production of the flange.

4.3.2 Machine Operations Performed to Produce the Final Part

The process of transforming the raw part into the final product involved several machine operations which were carried out in a precise and systematic manner.

The first operation was face milling, which involved the removal of excess raw material from the top and bottom surfaces of the part.

The next operation was profile milling which created a more defined profile, corresponding to the dimensions of the part, and also milled out the base of the part.

In addition, pocket milling was carried out to mill in the pockets on the shaft of the flange. The pockets were milled to a depth of 0.5 mm and had a width of 1 mm to ensure that they fit the intended purpose and provide an aesthetically pleasing appearance to the part.

Furthermore, a tapped hole was created for the central thread in the shaft of the flange. The hole was tapped to a diameter of 15 mm, a chamfer diameter of 16.5 mm, a core hole diameter of 13.1 mm, and a depth of 20 mm to accommodate the intended screw or bolt.

Lastly, a through-hole bore was made for the drills at the base of the flange. The holes were of a 2 mm diameter and a 5 mm depth.

All these machine operations were carried out with precision and accuracy to ensure that the final product met the design specifications and quality standards.

Once the part is transferred to EXAPTsolid, the production plan is created. The part's structure plan was created by considering one design parameter at a time. This approach was used to examine the effect of each parameter on the production process. After creating the structure plan for the first configuration, the part model was adjusted to create another configuration. By reloading the structure plan, a new corresponding CAM structure plan was automatically created.

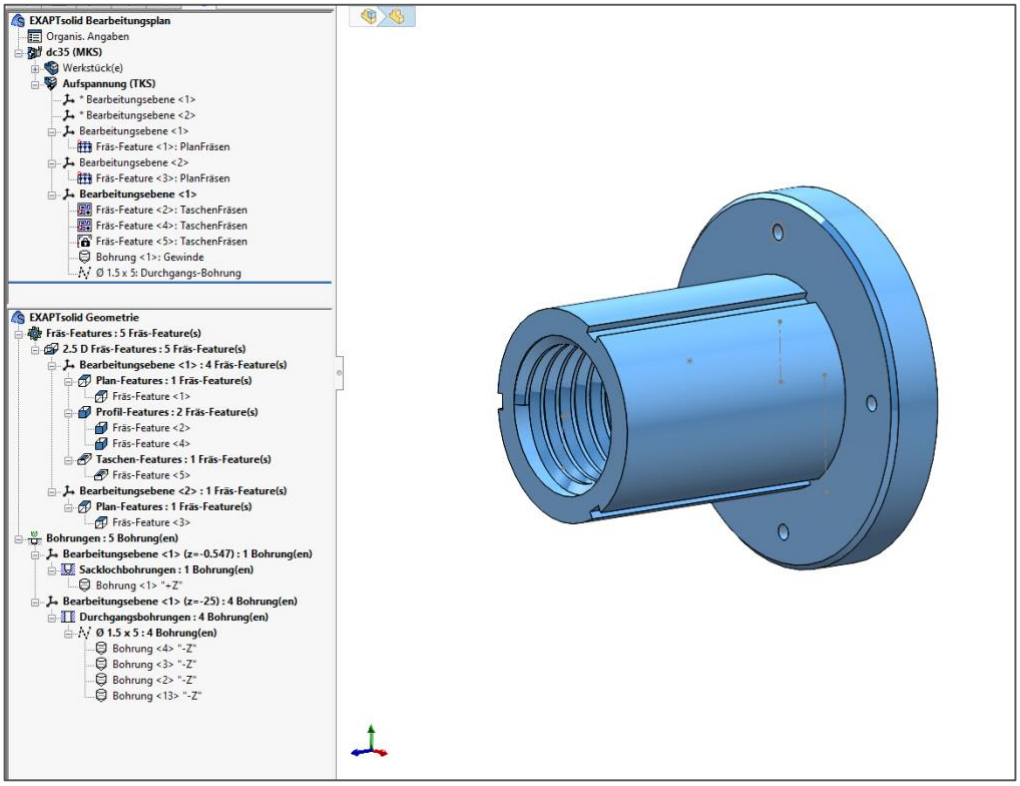


Figure 4: EXAPTsolid Structural Plan for Flange Configuration

In total, 26 configurations were created in this study for the 3 design parameters examined, with 10 for the number of holes on the base of the flange, 10 for the number of pockets on the shaft, and 6 for the length of the shaft. The creation of multiple configurations allowed for the examination of the impact of each design parameter on the production plan and overall manufacturability of the part.

4.4 Derivation of the Key Performance Indicators for the Analysis of the Machining Processes

KPIs, or Key Performance Indicators, are measurable values that are used to assess the effectiveness and success of a particular process, project, or organization. They serve as important metrics for monitoring progress towards specific goals and objectives and provide valuable insights into areas that may require improvement. KPIs are used in a wide range of industries and applications, and may be quantitative, such as production time, or qualitative.

The functions of KPIs include setting clear and measurable objectives, tracking progress towards these objectives, identifying areas for improvement, and ultimately driving performance and success. By monitoring and analysing KPIs, organizations can make data-driven decisions, allocate resources more effectively, and optimize their operations for maximum efficiency and profitability.

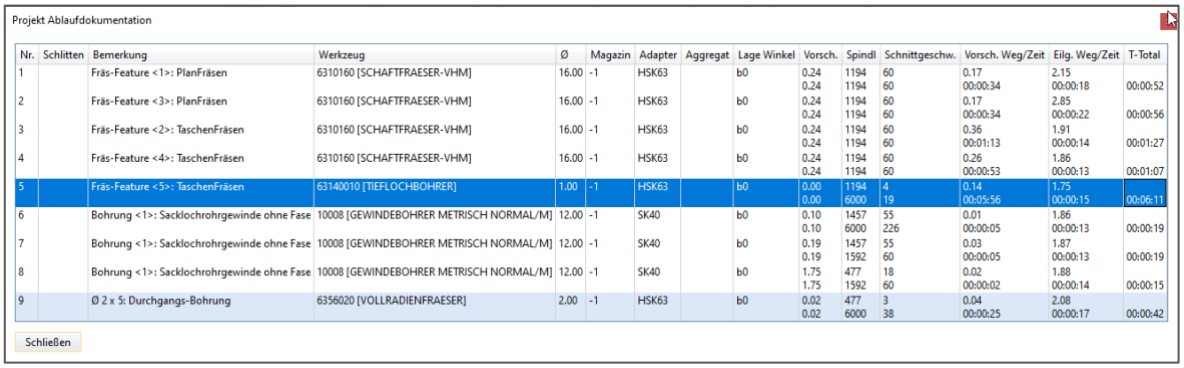
The following KPIs were generated to analyse the process for the production of the flange:

- **Tool Path in Feed Motion:** This KPI refers to the distance that the cutting tool covers while machining the part. It is used to evaluate the efficiency of the tool movement, which can directly impact the overall machining time.
- **Tool Path in Rapid Motion:** This KPI represents the distance covered by the cutting tool while moving at a higher speed without any material removal. It is used to determine the idle time of the tool and the time required for the tool to move between two machining points.
- **Total Time of Production:** This KPI is the total time taken for the start and completion of a machining object in the manufacture of the part.
- **Number of Tool Changes:** This KPI represents the total number of times tools were changed during the manufacturing process. It is used to evaluate the tool change strategy, which can have a significant impact on the machining time.
- **Maximum Diameter of Usable Cutting Tool:** This KPI represents the maximum diameter of the cutting tool that can be used for machining the part. It is used to evaluate the selection of the cutting tool, which can directly impact the efficiency of the machining process. For this part, the maximum diameter of the usable cutting tool was considered only for the manufacturing of the pockets.

5 Data Acquisition and Results

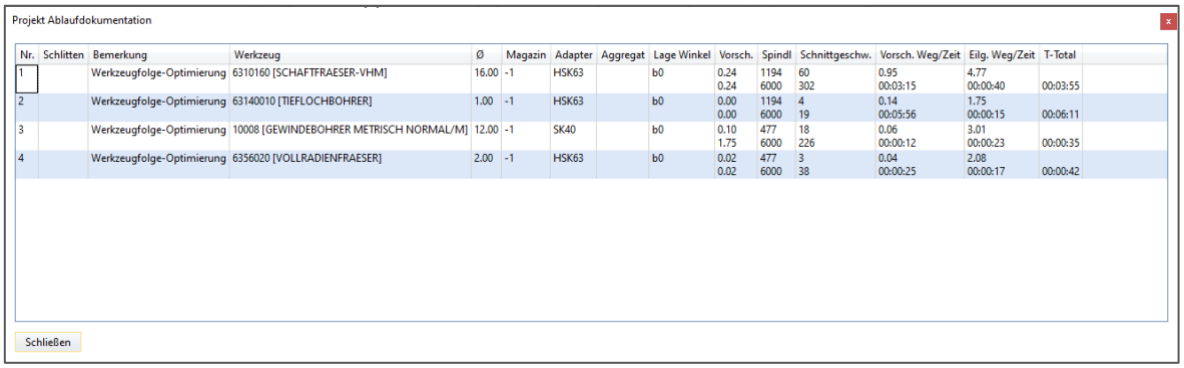
The process of data collection was facilitated and simplified by the utilisation of the EXAPTplus software. EXAPTplus provides a robust framework for flexible and efficient CAM programming in EXAPTsolid. This software tool operates in tandem with EXAPTsolid, offering a comprehensive range of features, including the EXAPT machine language. It contains all essential information about the various machining operations executed during CAM programming, including the key performance indicators (KPIs) examined in this study.

The data used for analysing the key performance indicators (KPIs) were obtained from two different sources. The data concerning toolpath length in feed and rapid motion, the number of tool changes, as well as the total time of production for each machining operation, was acquired from EXAPTplus under the Menu option: “Projektablaufdokumentation”. On the other hand, the data pertaining to the maximum radius of the usable cutting tool was directly extracted from EXAPTsolid.



Nr.	Schlitten	Bemerkung	Werkzeug	Ø	Magazin	Adapter	Aggregat	Lage Winkel	Vorsch.	Spindl	Schnittgeschw.	Vorsch. Weg/Zeit	Eilg. Weg/Zeit	T-Total
1		Fräs-Feature <1>: PlanFräsen	6310160 [SCHAFTFRAESER-VHM]	16.00	-1	HSK63		b0	0.24 0.24	1194 1194	60 60	0.17 00:00:34	2.15 00:00:18	00:00:52
2		Fräs-Feature <3>: PlanFräsen	6310160 [SCHAFTFRAESER-VHM]	16.00	-1	HSK63		b0	0.24 0.24	1194 1194	60 60	0.17 00:00:34	2.85 00:00:22	00:00:56
3		Fräs-Feature <2>: TaschenFräsen	6310160 [SCHAFTFRAESER-VHM]	16.00	-1	HSK63		b0	0.24 0.24	1194 1194	60 60	0.36 00:01:13	1.91 00:00:14	00:01:27
4		Fräs-Feature <4>: TaschenFräsen	6310160 [SCHAFTFRAESER-VHM]	16.00	-1	HSK63		b0	0.24 0.24	1194 1194	60 60	0.26 00:00:53	1.86 00:00:13	00:01:07
5		Fräs-Feature <5>: TaschenFräsen	63140010 [TIEFLOCHBOHRER]	1.00	-1	HSK63		b0	0.00 0.00	1194 6000	4 19	0.14 00:05:56	1.75 00:00:15	00:06:11
6		Bohrung <1>: Sacklochrohrgewinde ohne Fase	10008 [GEWINDEBOHRER METRISCH NORMAL/M]	12.00	-1	SK40		b0	0.10 0.10	1457 6000	55 226	0.01 00:00:05	1.86 00:00:13	00:00:19
7		Bohrung <1>: Sacklochrohrgewinde ohne Fase	10008 [GEWINDEBOHRER METRISCH NORMAL/M]	12.00	-1	SK40		b0	0.19 0.19	1457 1592	55 60	0.03 00:00:05	1.87 00:00:13	00:00:19
8		Bohrung <1>: Sacklochrohrgewinde ohne Fase	10008 [GEWINDEBOHRER METRISCH NORMAL/M]	12.00	-1	SK40		b0	1.75 1.75	477 1592	18 60	0.02 00:00:02	1.88 00:00:14	00:00:15
9		Ø 2 x 5: Durchgangs-Bohrung	6356020 [VOLLRADIENFRAESER]	2.00	-1	HSK63		b0	0.02 0.02	477 6000	3 38	0.04 00:00:25	2.08 00:00:17	00:00:42

Figure 5: Data Representation on EXAPTplus



Nr.	Schlitten	Bemerkung	Werkzeug	Ø	Magazin	Adapter	Aggregat	Lage Winkel	Vorsch.	Spindl	Schnittgeschw.	Vorsch. Weg/Zeit	Eilg. Weg/Zeit	T-Total
1		Werkzeugfolge-Optimierung	6310160 [SCHAFTFRAESER-VHM]	16.00	-1	HSK63		b0	0.24 0.24	1194 6000	60 302	0.95 00:03:15	4.77 00:00:40	00:03:55
2		Werkzeugfolge-Optimierung	63140010 [TIEFLOCHBOHRER]	1.00	-1	HSK63		b0	0.00 0.00	1194 6000	4 19	0.14 00:05:56	1.75 00:00:15	00:06:11
3		Werkzeugfolge-Optimierung	10008 [GEWINDEBOHRER METRISCH NORMAL/M]	12.00	-1	SK40		b0	0.10 1.75	477 6000	18 226	0.06 00:00:12	3.01 00:00:23	00:00:35
4		Werkzeugfolge-Optimierung	6356020 [VOLLRADIENFRAESER]	2.00	-1	HSK63		b0	0.02 0.02	477 6000	3 38	0.04 00:00:25	2.08 00:00:17	00:00:42

Figure 6: Data Representation on EXAPTplus Emphasizing the Number of Tool Changes

The data obtained for each machining operation on each variation of the part was recorded and graphically depicted in accordance with the parameters that were analysed.

5.1 Variation in the Number of Holes

10 different variations were examined under the parameter "number of holes" with the number ranging from 3 to 12 holes. The diameter of each hole was kept constant at 2mm, and the spacing between the holes was also maintained equally. This was done to ensure that any force applied on the flange would be transferred equally through its base.

During the analysis of the variations for this parameter, the KPIs considered included tool path length during both feed and rapid motion, total production time per machining object, and the number of tool changes required during the machining process.

Number of Holes	Toolpath length in feed motion in m	Toolpath length in rapid motion in m	Time of Production in s	Number of Tool Changes
3.00	0.03	2.08	35.00	4.00
4.00	0.04	2.08	42.00	4.00
5.00	0.05	2.11	49.00	4.00
6.00	0.06	2.12	56.00	4.00
7.00	0.07	2.15	63.00	4.00
8.00	0.08	2.14	70.00	4.00
9.00	0.09	2.15	77.00	4.00
10.00	0.10	2.17	83.00	4.00
11.00	0.11	2.18	90.00	4.00
12.00	0.12	2.20	97.00	4.00

Table 2: Data Collection - Variation in number of holes

After carrying out the process, the following results were obtained.

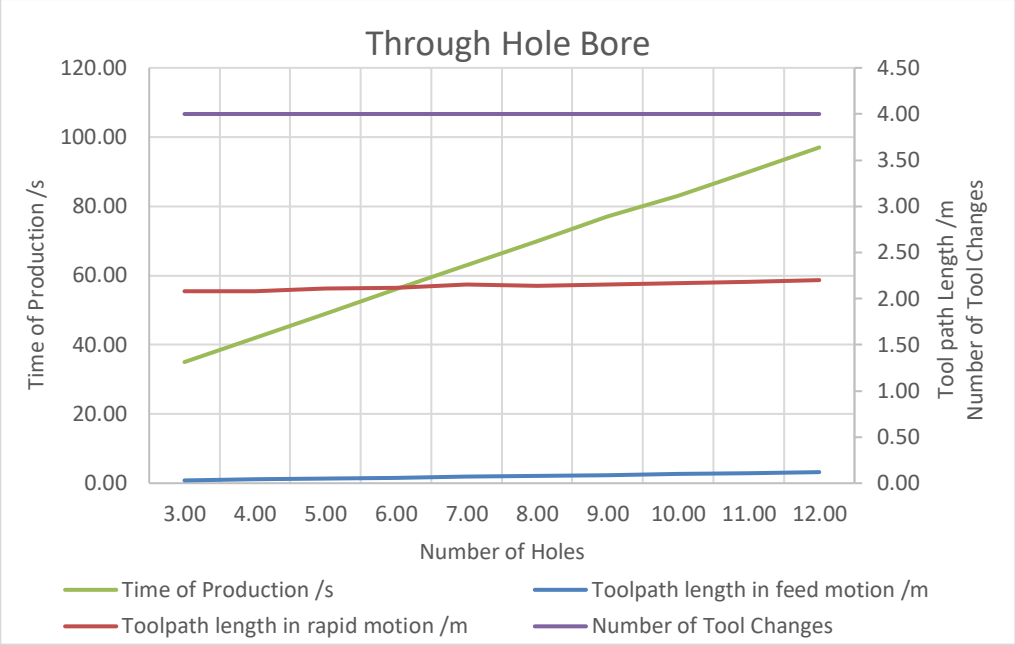


Figure 7: Graphical Representation: Variation in the Number of Holes

The total tool path length increased with an increase in the number of holes, with the maximum length being recorded when the flange had 12 holes. Similarly, the total production time increased with an increase in the number of holes, with the longest production time being recorded for the flange with 12 holes. However, the number of tool changes remained constant for all the variations since the structure plan was automatically updated.

5.2 Number of Pockets

Similarly, the flange was examined under the parameter "number of pockets on the shaft." A total of 10 variations were analysed, with the number of pockets ranging from 3 to 12. The pockets were all of constant diameter of 1mm and depth of 0.5 mm, with equal spacing between one another.

The KPIs analysed included the tool path length in both feed and rapid motion, the total production time per machining object, the maximum diameter of the cutting tool for the pockets, and the number of tool changes required.

Number of Pockets	Toolpath length in feed motion in m	Toolpath length in rapid motion in m	Time of Production in s	Number of Tool Changes	Max Diameter of Cutting tool in mm
3.00	0.11	1.71	273.00	4.00	1.00
4.00	0.14	1.75	360.00	4.00	1.00
5.00	0.18	1.79	447.00	4.00	1.00
6.00	0.21	1.82	534.00	4.00	1.00
7.00	0.25	1.87	621.00	4.00	1.00
8.00	0.29	1.89	708.00	4.00	1.00
9.00	0.32	1.89	795.00	4.00	1.00
10.00	0.75	1.62	1831.00	4.00	1.00
11.00	0.75	1.62	1894.00	4.00	1.00
12.00	0.43	1.97	1090.00	4.00	1.00

Table 3: Variation in the Number of Pockets - Data Collection

After plotting the data, the following results were obtained.

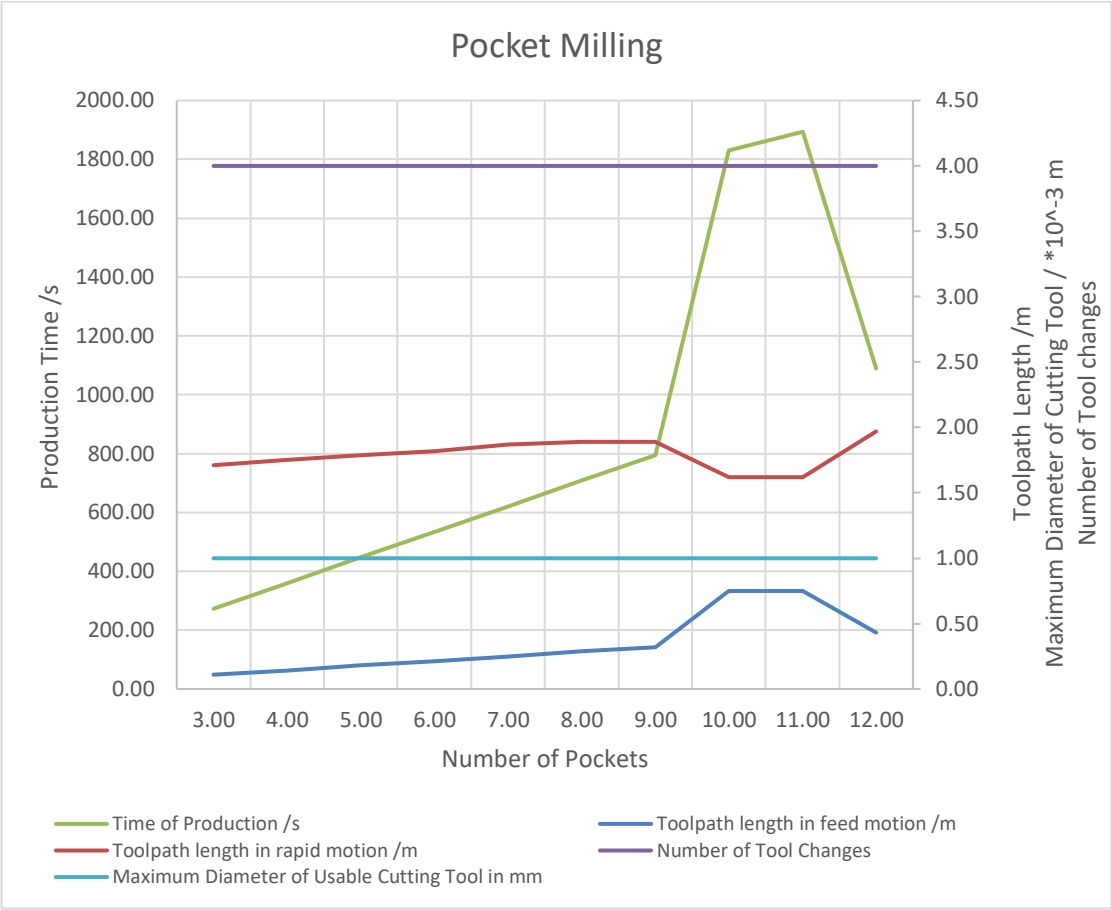


Figure 8: Graphical Representation - Variation in the Number of Pockets

The tool path length in feed motion ranged from 0.11 m for three pockets, over 0.75 m for 10 and 11 pockets, to 0.43 m for twelve pockets. The tool path length in rapid motion ranged from 1.71 m for three pockets to 1.97 m for twelve pockets, with a slight plummet at 1,62 for 10 and 11 pockets. The total production time per machining object ranged from 273 seconds for three pockets to 1900 seconds for 11 pockets, sinking again to 1090 seconds for 12 pockets. The maximum diameter of the cutting tool required for the pockets remained constant at 1 mm for all variations, since the same tool was used for the milling of each pocket. Finally, the number of tool changes required for each variation also remained constant at 4.

5.3 Length of the Shaft

The production of the flange's shaft required three specific machining operations, namely, end milling on the top surface of the flange, as well as two profile milling features for the cylindrical profile of the shaft. In order to examine the effects of different shaft lengths, the length of the shaft was varied between 5 mm and 30 mm, in intervals of 5 mm. As a result, six variations were analysed under this parameter.

The KPIs measured included tool path length in both feed and rapid motion, total production time per machining object, and the number of tool changes required to complete the manufacturing process.

- **End Milling – Top of the Flange**

Length of Shaft in mm	Toolpath length in feed motion in m	Toolpath length in rapid motion in m	Time of Production in s	Number of Tool Changes
5.00	0.17	2.27	53.00	4.00
10.00	0.17	2.27	53.00	4.00
15.00	0.17	2.21	53.00	4.00
20.00	0.17	2.18	53.00	4.00
25.00	0.17	2.15	52.00	4.00
30.00	0.17	2.12	52.00	4.00

Table 4: Data Collection - End Milling for the Top of the Flange

- **Profile Milling – Flange Base Contour**

Length of Shaft mm	Toolpath length in feed motion in m	Toolpath length in rapid motion in m	Time of Production in s	Number of Tool Changes
5.00	0.34	1.93	83.00	4.00
10.00	0.34	1.93	84.00	4.00
15.00	0.35	1.92	85.00	4.00
20.00	0.35	1.92	86.00	4.00
25.00	0.36	1.91	87.00	4.00
30.00	0.69	1.98	157.00	4.00

Table 5: Data Collection - Profile Milling for the Base of the Flange

- **Profile Milling – Flange Shaft Contour**

Length of Shaft in mm	Toolpath length in feed motion in m	Toolpath length in rapid motion in m	Time of Production in s	Number of Tool Changes
5.00	0.21	1.93	54.00	4.00
10.00	0.21	1.93	55.00	4.00
15.00	0.22	1.92	56.00	4.00
20.00	0.43	1.92	57.00	4.00
25.00	0.43	1.97	99.00	4.00
30.00	0.43	1.96	100.00	4.00

Table 6: Data Collection - Profile Milling for the Shaft of the Flange

After conducting the analysis, the following results were obtained for the six variations of shaft length.

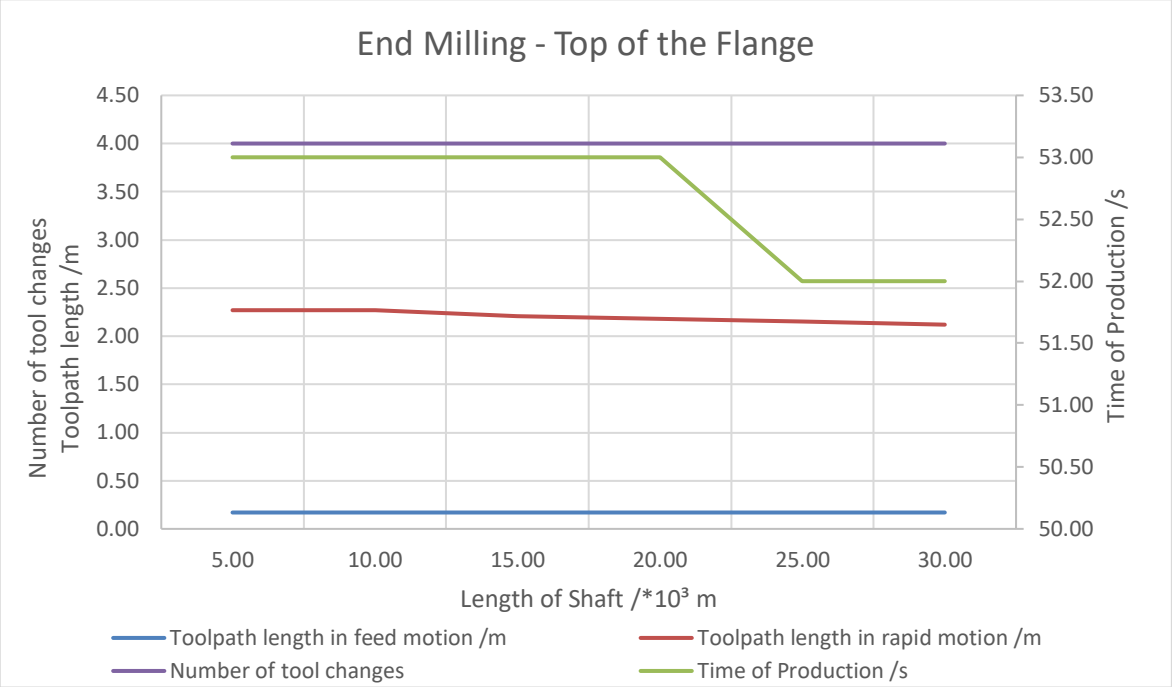


Figure 9: Graphical Representation - End Milling of the Top of the Flange

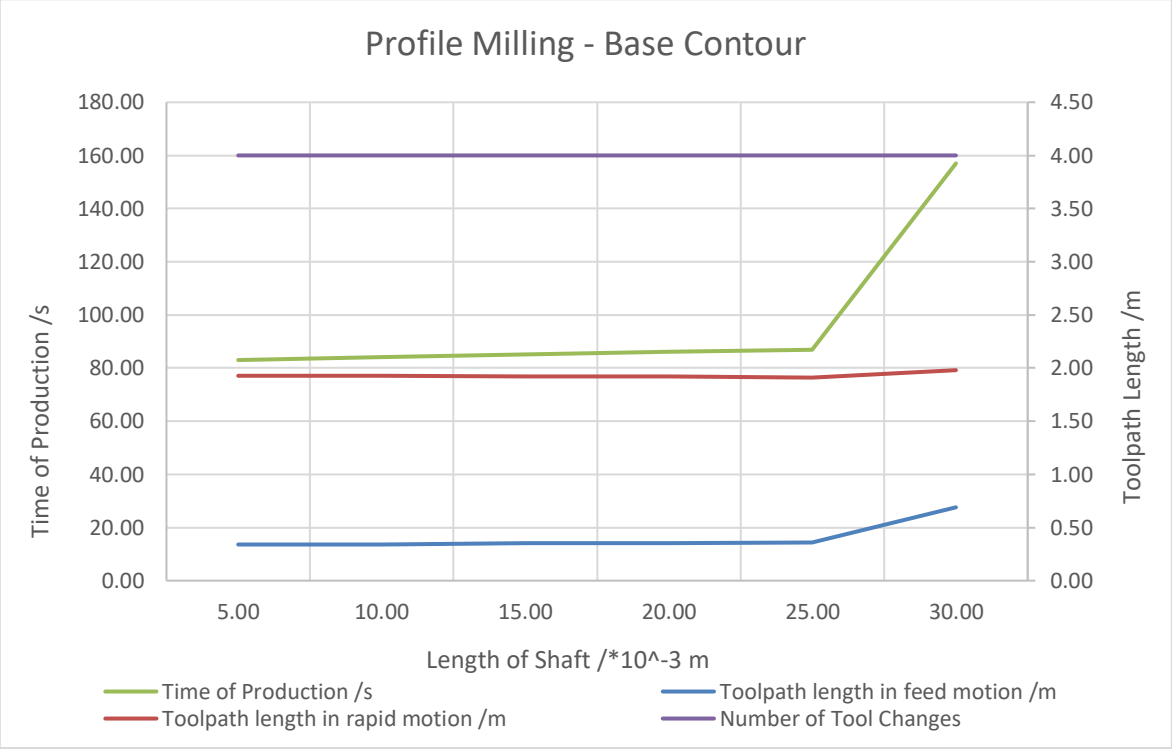


Figure 10: Graphical Representation - Base Contour Milling

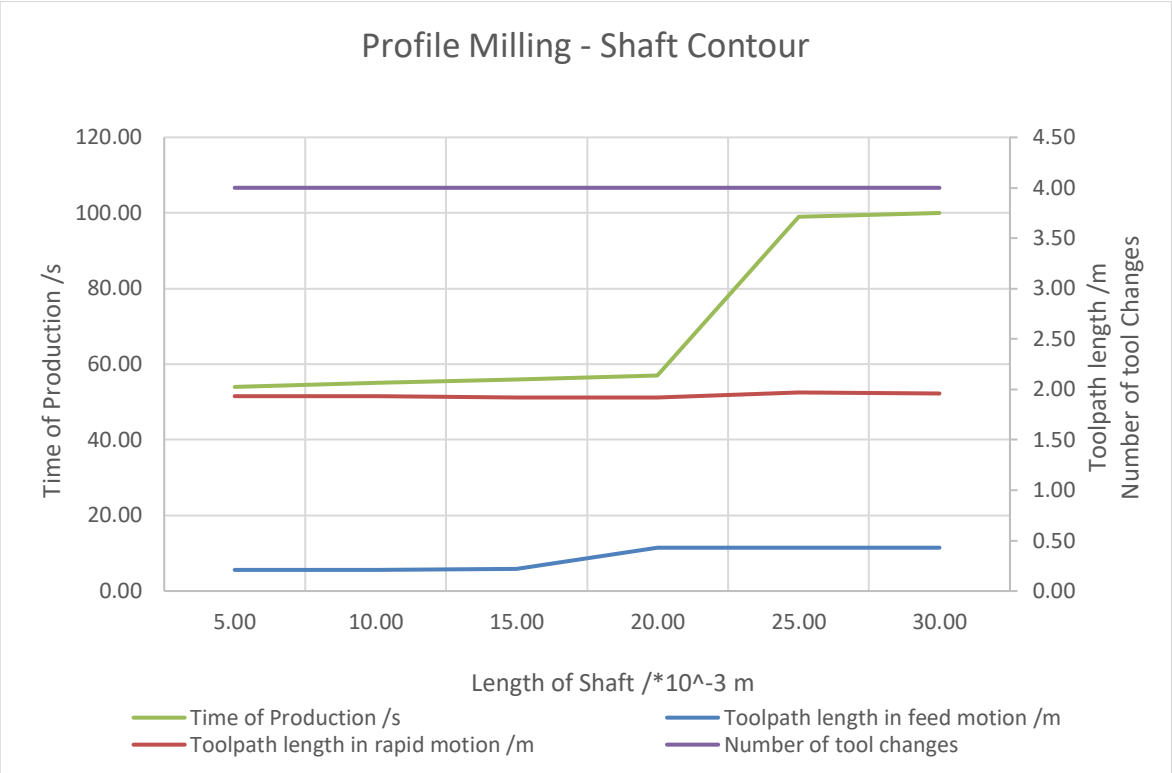


Figure 11: Graphical Representation - Shaft Contour Milling

In terms of tool path length in feed motion for the end milling, there was no great change in the variations, as the toolpath in feed motion remained at 0.17 m. However, it ranged between 0.34 m to 0.69 m for the base contour profile milling and ranged between 0.21 m and 0.43 m for the shaft contour profile milling, from the shortest to the longest shaft. On the other hand, the variation with the shortest shaft length had the highest tool path length in rapid motion at 2.27 m, while the variation with the longest shaft length had the lowest tool path length at 2.12 m.

In terms of the total production time per machining object, the analysed variations exhibited a range of production times. End milling showed a slow decrease in production time from 53 to 52 seconds, while the base contour displayed a significant increase, ranging from 83 to 157 seconds. Additionally, the production time for the shaft contour varied between 54 and 100 seconds.

It is worth noting that due to the automatic update of the structure plan from one configuration to another, the number of tool changes remained constant at 4 for the total manufacturing of the tool.

6 Evaluation of the Part and Guideline Proposal

6.1 Evaluation of the CAD Parameters and Proposal of Guidelines

Based on the results obtained from the analysis of the KPIs of the machining process, the following guidelines have been proposed to optimize the manufacturing of the flange. These guidelines cover various aspects of the machining process, including pocket diameter, pocket length, pocket fillet, shaft length and hole diameter, and they are expected to enhance the manufacturability of the flange and improve the quality of the final product through the optimisation of the KPIs.

- **Make the diameter of the pockets identical to that of the holes:**
  By ensuring that the diameter of the pockets matches that of the holes, the process of machining the flange can be streamlined. This would minimize the amount of time spent on tool changeovers and reduce the total production time.
- **Fillet the inner angles of the pockets of the flange:**
  By adding fillets to the inner angles of the pockets, the range of the maximum diameter of the usable cutting tool is increased. This would enable a wider range of tools to be used in the machining process, making the process more efficient and cost-effective.
- **Reduce the length of the shaft:**
  Additionally, reducing the length of the shaft would decrease the amount of material needed, resulting in cost savings during manufacturing.
- **Match the length of the pocket to that of the shaft:**
  By matching the length of the pocket to that of the shaft, the process of machining the flange can be further optimized. This would reduce the total tool path length and minimize the amount of time spent on tool changeovers.

6.2 Implementation of the Guidelines in the CAD Model

The implementation of the proposed guidelines in the design of the part involved several adjustments made using SolidWorks. The main focus of these adjustments was on the pockets of the flange, where changes were made to align the diameter of the pockets with that of the holes (d = 2 mm). Additionally, modifications were made to the shaft of the flange as per the proposed guidelines. These adjustments and refinements resulted in a final product that exhibits a design where the pockets and holes are appropriately aligned, ensuring optimal functionality and enhanced manufacturing efficiency. The final model reflects the successful integration of the proposed guidelines into the design process, yielding a visually refined and functionally optimized flange.

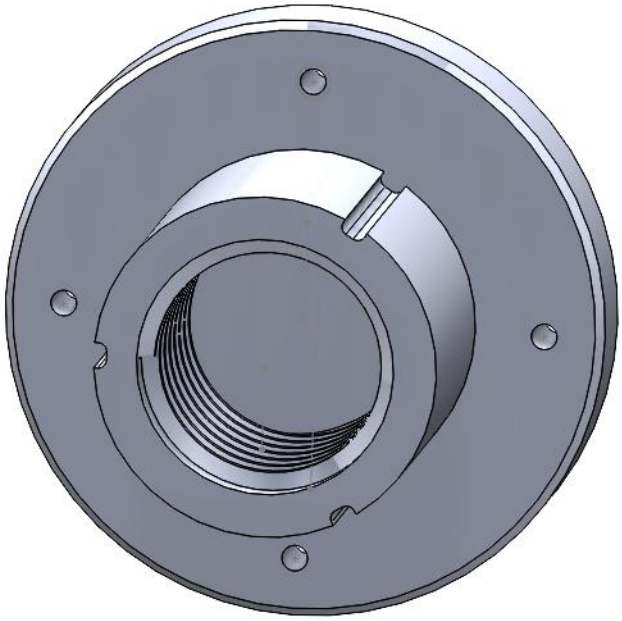


Figure 12: Design after Guideline Implementation

6.3 Implementation of Guidelines on the Machining Operations

After implementing the guidelines on the part, corresponding changes had to be made to the machining operations for better optimization. The changes involved merging some machining operations and using the same tool for multiple operations, which ultimately reduced the number of tool changes needed for the production of the flange. For instance, Tool ID 6310160 and Tool ID 63100100, which were previously used for the single machining operation of the milling profile of the shaft and base of the flange, were replaced by Tool ID 6310160, which could perform all four machining operations. As a result, the total production time was significantly reduced.

Moreover, to optimize the machining further, Tool ID 631410 and Tool ID 6356020, responsible for milling the pockets and boring the holes, respectively, were replaced with Tool 6310020, which could perform both operations while adhering to both the proposed guidelines and the general engineering norms.

The table below provides an overview of all the tools used for the machining operations.

Tool ID	Tool Name	Function
6310160	End mill	Used for making shafts, shapes, and holes on the workpiece
6310020	End mill	Used for making shafts, shapes, and holes on the workpiece
10008	Tapping Tool (metric normal)	Used for creating internal threads in the workpiece

Table 7: Tools and Corresponding IDs in EXAPT Database

The machining operations were executed on various variations of the part while varying the three principal parameters: the number of pockets, length of the shaft, and number of holes. Following this, the data was collected and subjected to a re-analysis.

6.4 Data Acquisition and Results after Implementation of the Proposed Guidelines

This section presents the data collected and the results obtained after the implementation of the proposed guidelines. The data collected during this machining process will be analysed and compared to the data collected before the implementation of the guidelines. The results are expected to provide insight into the impact of the guidelines on the machining process and highlight the areas where improvements were observed. The average percentage difference between the data to determine the effectiveness of the guidelines in enhancing machining production efficiency is calculated using the formula:

$$Average\ Percentage\ Difference = \frac{\sum\left(\left(\frac{x_{before} - x_{after}}{x_{before}}\right) \cdot 100\right)}{Number\ of\ Variations} \quad (1)$$

Where $x_{before} = value\ of\ the\ data\ collected\ before\ implementation\ of\ the\ guidelines$ and $x_{after} = value\ of\ the\ data\ collected\ before\ implementation\ of\ the\ guidelines.$

6.4.1 Number of Holes

In order to investigate the effect of the guidelines on the machining process, a total of 10 different variations were examined, with the number of holes ranging from 3 to 12. The diameter of the holes was kept constant at 2mm for all variations. However, a different tool with ID 6310020 was used for this investigation. The collected data was then compiled and analysed, and the results are presented in the following table.

Number of Holes	Toolpath length in feed motion – After / m	Toolpath length in rapid motion – After in m	Time of Production - After in s	Number of tool changes
3.00	0.03	2.18	31.00	2.00
4.00	0.04	2.14	37.00	2.00
5.00	0.05	2.18	43.00	2.00
6.00	0.05	2.18	49.00	2.00
7.00	0.06	2.19	54.00	2.00
8.00	0.07	2.22	60.00	2.00
9.00	0.08	2.20	66.00	2.00
10.00	0.09	2.23	71.00	2.00
11.00	0.10	2.23	77.00	2.00
12.00	0.11	2.25	82.00	2.00

Table 8: Data Collection - Variation in the Number of Holes after Implementation

The KPIs were analysed separately to compare the performance before and after the implementation of the guidelines. The KPIs that were evaluated include the number of tool changes, the total time of production, the toolpath in feed motion, and the toolpath in rapid motion.

After conducting the analysis, the following results were obtained for the 10 variations of the number of pockets.

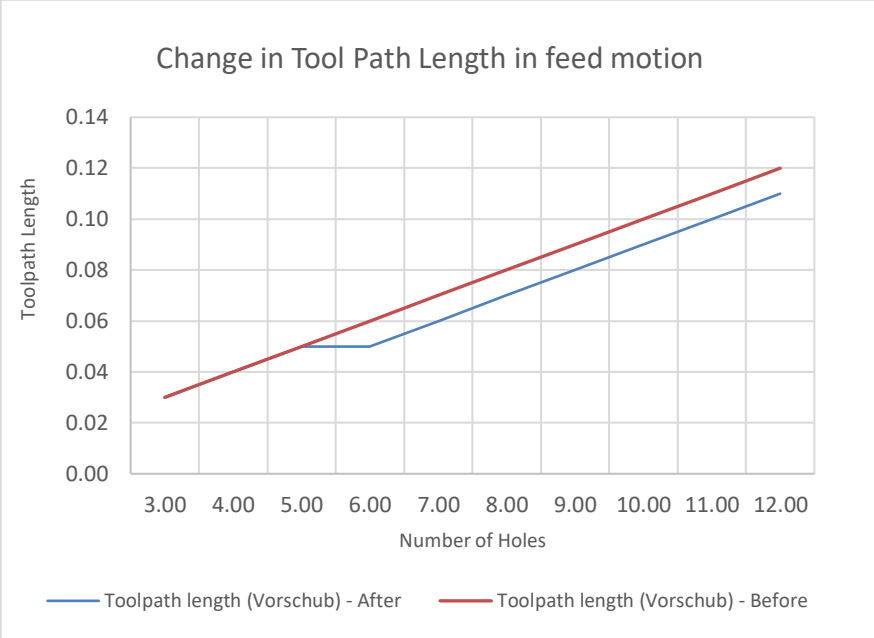


Figure 13: Change in the Toolpath Length in Feed Motion

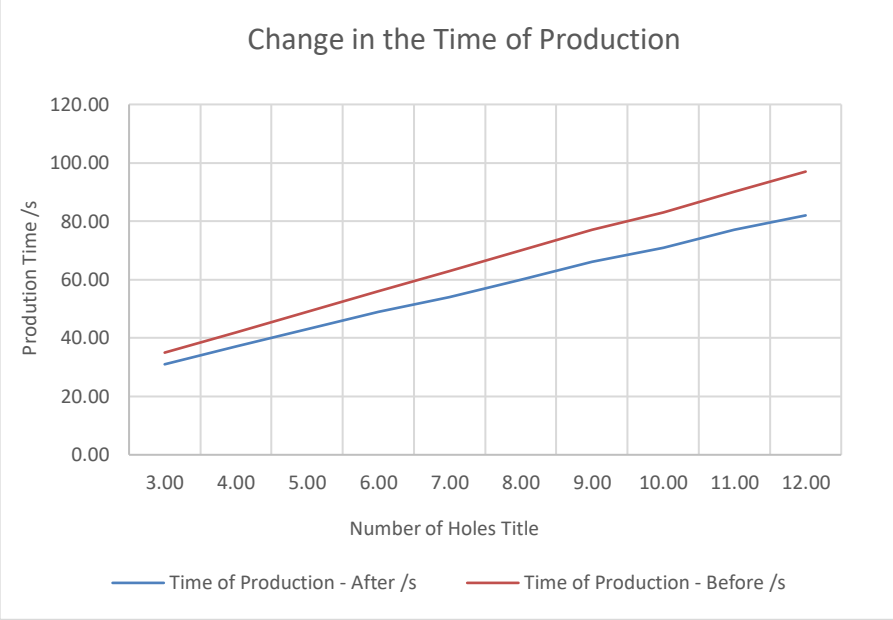


Figure 14: Change in Production Time

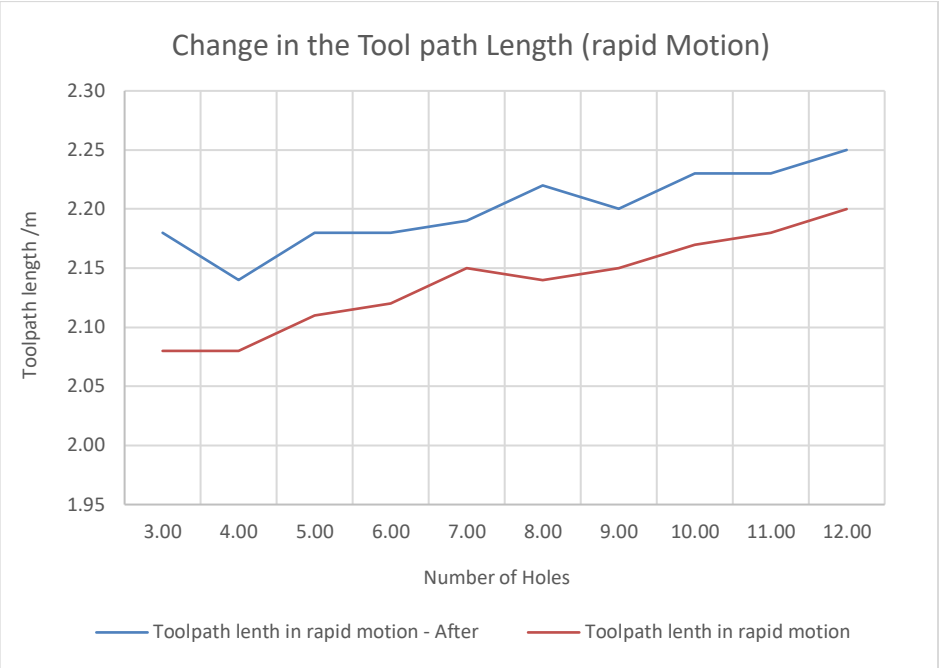


Figure 15: Change in the Toolpath Length in Rapid Motion

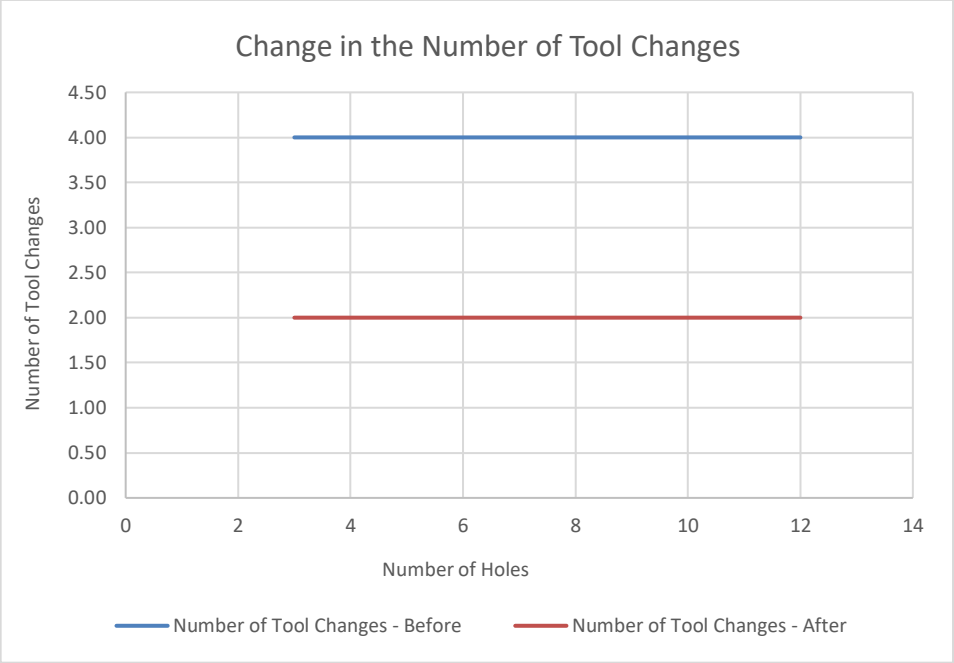


Figure 16: Change in the Number of Tool Changes

The results indicated a significant improvement in the performance of the machining operations. The total production time was reduced by 13,5%, the number of tool changes by 50% and the toolpath length in feed motion decreased by 8,2%. However, the implementation of the guidelines resulted in an increase in the toolpath length in rapid motion by 2,9% as compared to before the implementation. This change may be attributed to the substitution of the tool used to bore, which could have affected the machining path and led to the observed increase in toolpath length during rapid motion.

6.4.2 Number of Pockets

Similarly, to the first parameter, the machining operations were conducted on 10 different variations under the second parameter "number of pockets" with the number ranging from 3 to 12 pockets. The machining operations were carried out after following the guidelines which included matching the diameter of the pockets to that of the holes (2mm) to reduce the number of tool changes in producing the part. After the implementation of the guidelines, the same tool (Tool ID 6310020) was used to generate the pockets.

The following data was collected.

Number of Pockets	Toolpath length in feed motion - After in m	Toolpath length in rapid motion – After in m	Time of Production - After in s	Number of tool changes	Max Diameter of Cutting tool in mm
3.00	0.11	2.19	76.00	2.00	2.00
4.00	0.14	2.18	96.00	2.00	2.00
5.00	0.18	2.21	117.00	2.00	2.00
6.00	0.21	2.25	137.00	2.00	2.00
7.00	0.25	2.27	158.00	2.00	2.00
8.00	0.27	2.02	164.00	2.00	2.00
9.00	0.57	2.07	331.00	2.00	2.00
10.00	0.36	2.38	220.00	2.00	2.00
11.00	0.39	2.37	240.00	2.00	2.00
12.00	0.27	2.01	165.00	2.00	2.00

Table 9: Data Collection - Variation in the Number of Pockets after Implementation

The KPIs that were taken into consideration include the number of tool changes, total production time, toolpath in feed motion, maximum diameter of the usable cutting tool, and toolpath in rapid motion. The results obtained from the analysis of the KPIs for these 10 variations are presented below.

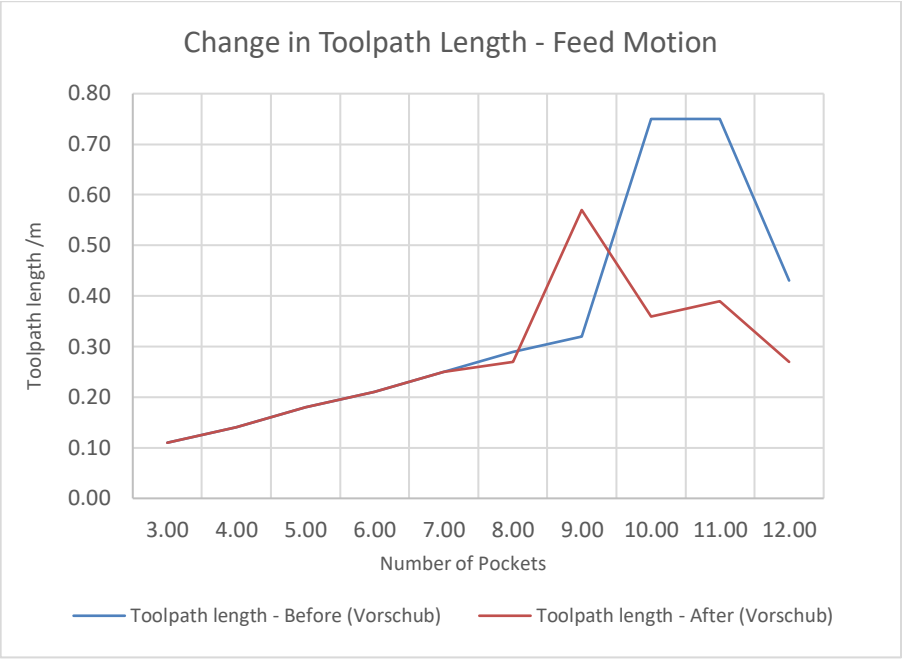


Figure 17: Change in the Toolpath Length in Feed Motion



Figure 18: Change in Production Time

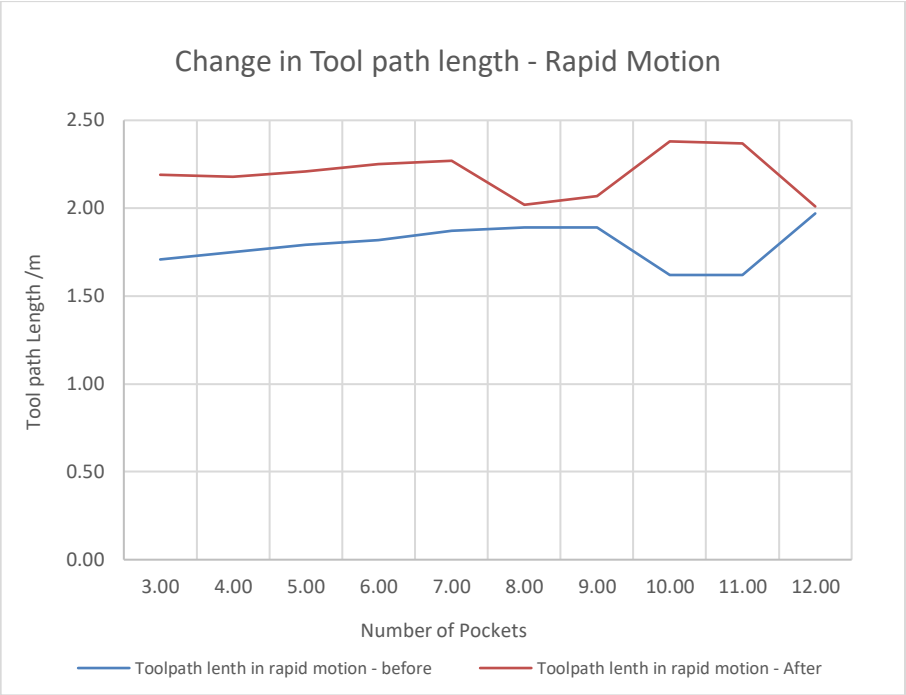


Figure 19: Change in the Toolpath Length in Rapid Motion

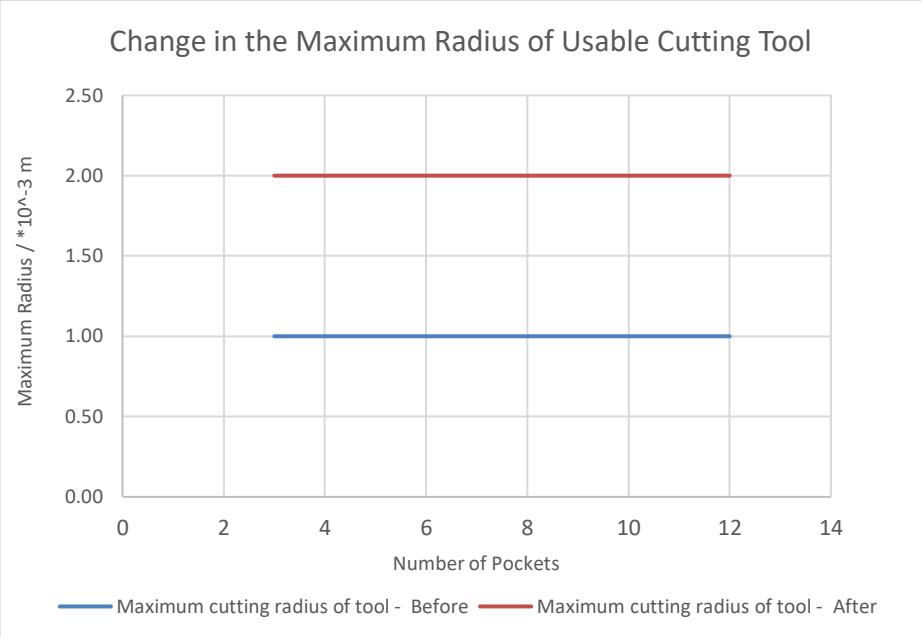


Figure 20: Change in the Maximum Radius of Cutting Tool

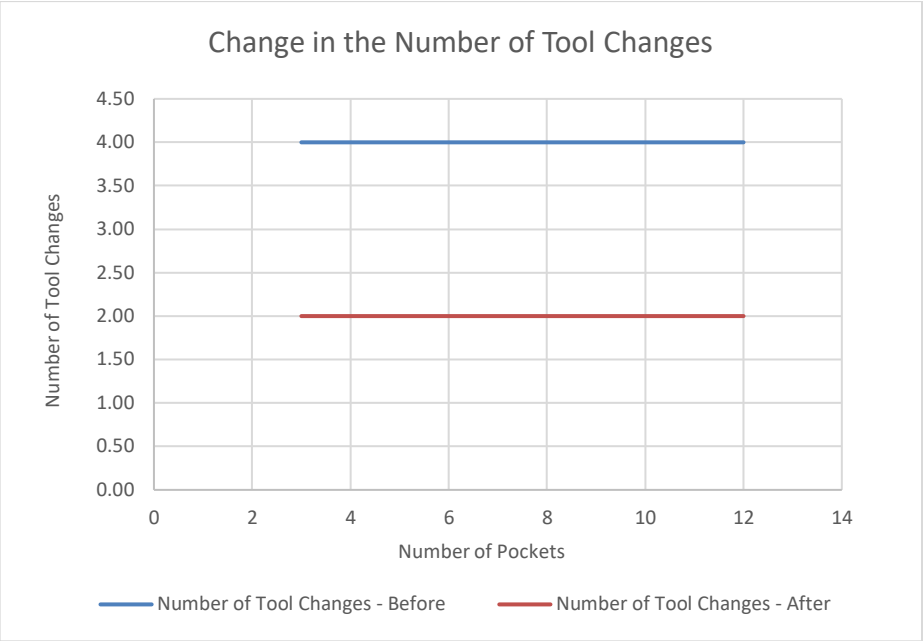


Figure 21: Change in the Number of Tool Changes

Upon analysing the data obtained before and after implementing the guidelines using graphical representations, the findings revealed a marginal decrease by 6.6% in the toolpath length during feed motion. Moreover, a significant reduction of 76.4% in the total production time was also observed.

However, a noticeable 23.3% increase in the toolpath length during rapid motion was detected, indicating a potential deviation from the expected outcome. As aforementioned, this outcome might have been due to the change in the tool, which could have caused the difference in the observed tool path.

6.4.3 Length of Shaft

In a similar fashion to the previous parameter analyses, the length of the shaft was varied between 5 mm and 30 mm, in increments of 5 mm, resulting in six variations being examined. The KPIs examined included tool path length in both feed and rapid motion, total production time per machining object, and the time required to complete the manufacturing.

- **End Milling – Top of the Flange**

Length in mm	Toolpath length in feed motion - After in m	Toolpath length in rapid motion – After in m	Time of Production - After in s	Number of Tool Changes - After
5.00	0.17	2.23	53.00	2.00
10.00	0.17	2.21	53.00	2.00
15.00	0.17	2.19	53.00	2.00
20.00	0.17	2.17	53.00	2.00
25.00	0.17	2.15	53.00	2.00
30.00	0.17	2.19	53.00	2.00

Table 10: Data Collection – End Milling for the Top of the Flange

Profile Milling – Base Contour

Length in mm	Toolpath length in feed motion -After	Toolpath length in rapid motion - After	Time of Production - After in s	Number of Tool Changes - After
5.00	0.34	1.89	83.00	2.00
10.00	0.34	1.90	84.00	2.00
15.00	0.35	1.90	85.00	2.00
20.00	0.35	1.91	86.00	2.00
25.00	0.36	1.91	87.00	2.00
30.00	0.69	1.99	157.00	2.00

Table 11: Data Collection – Profile Milling for the Base of the Flange

- **Profile Milling – Shaft Contour**

Length in mm	Toolpath length in feed motion - After	Toolpath length in rapid motion - After	Time of Production - After in s	Number of Tool Changes - After
5.00	0.24	1.84	62.00	2.00
10.00	0.24	1.85	63.00	2.00
15.00	0.25	1.85	64.00	2.00
20.00	0.25	1.86	65.00	2.00
25.00	0.26	1.86	67.00	2.00
30.00	0.26	1.87	68.00	2.00

Table 12: Data Collection – Profile Milling for the Shaft of the Flange

By comparing the KPIs before and after implementing the guidelines, the impact of the changes on the machining process resulted in ^the following graphical representations.

- **End Milling – Top of the Flange**

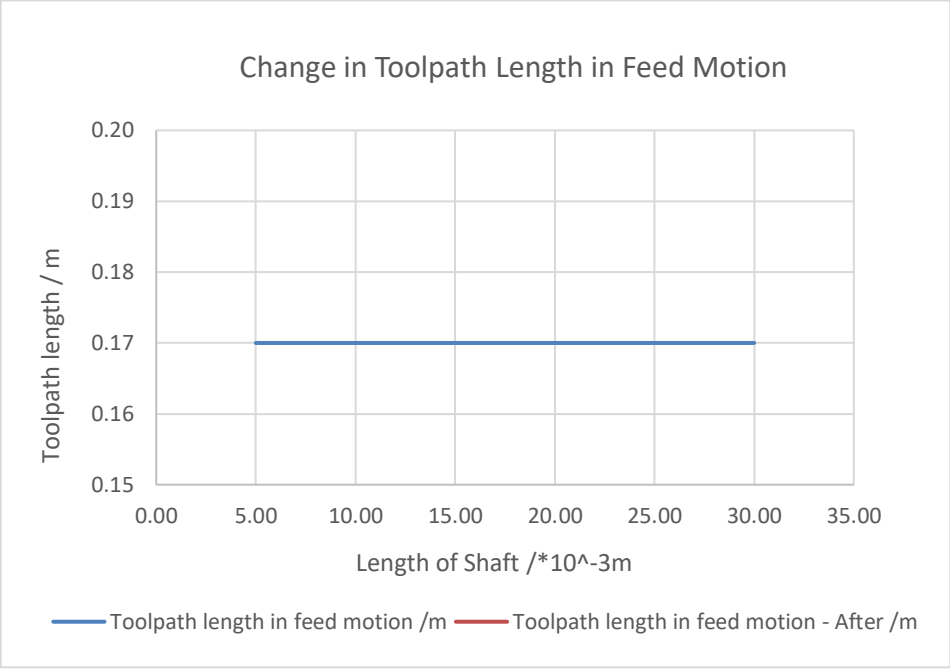


Figure 22: Change in the Toolpath Length in Feed Motion

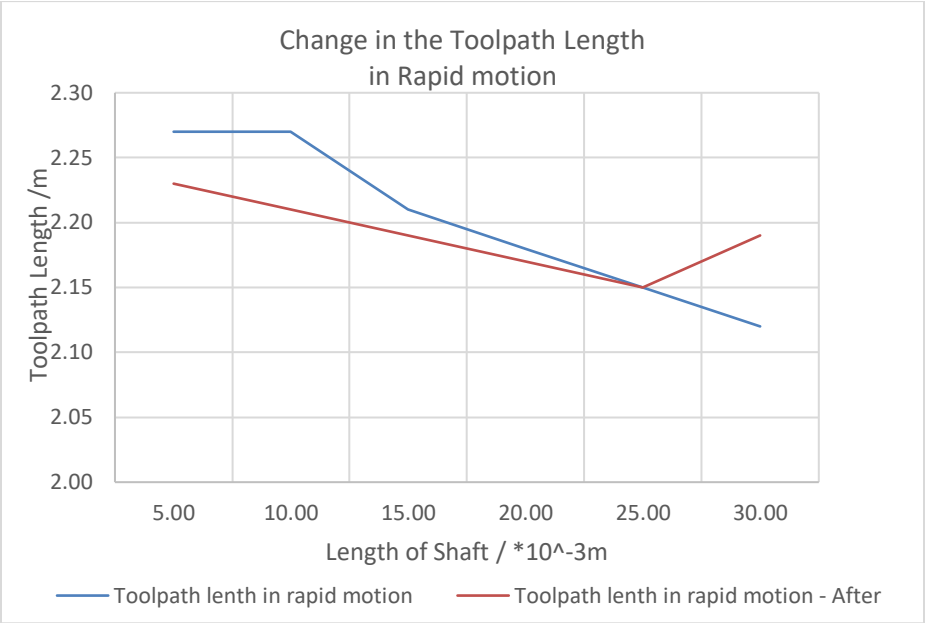


Figure 23: Change in the Toolpath Length in Rapid Motion

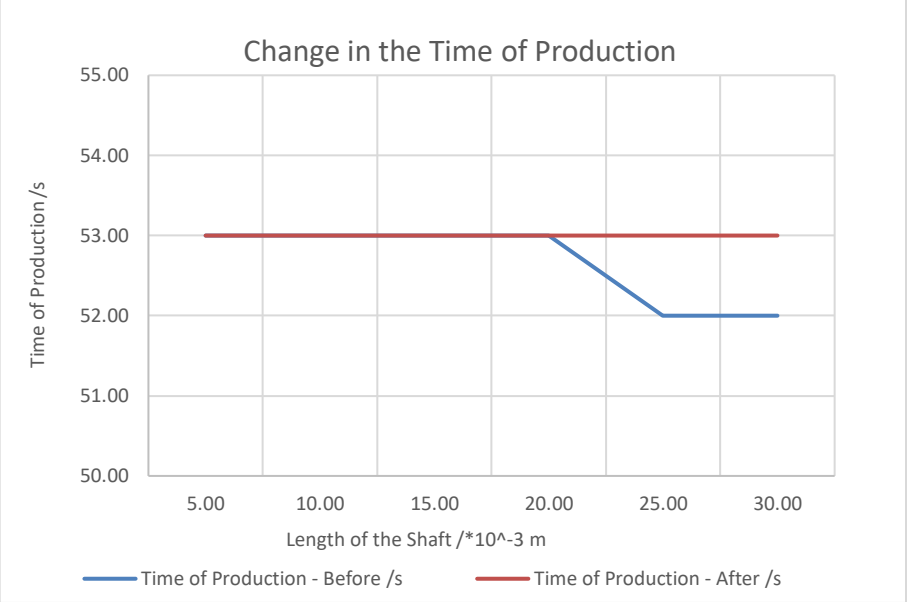


Figure 24: Change in the Production Time

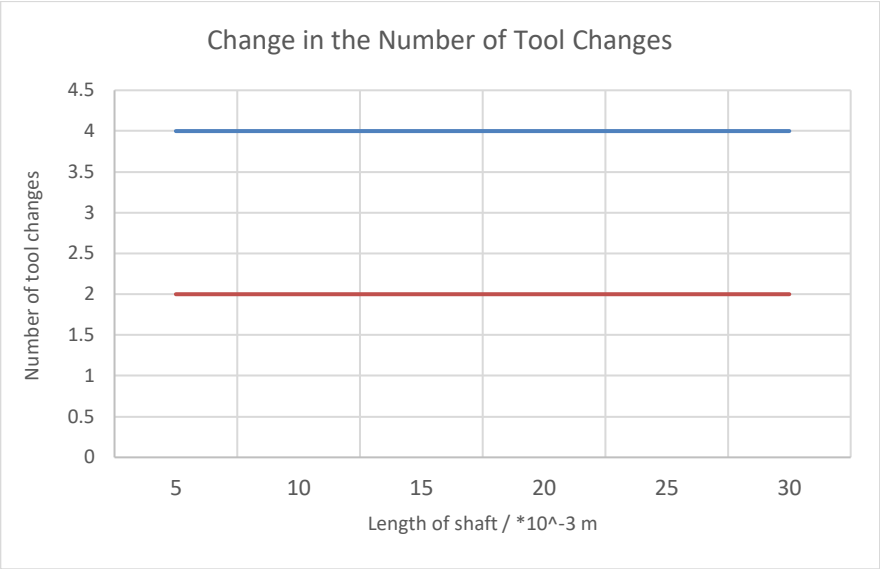


Figure 25: Change in the Number of Tool Changes

In the case of end milling the top surface of the flange, minimal changes were observed in the graphical representation. The tool path in feed motion remained consistently stable, exhibiting no significant variations. However, a marginal increase of 0.6% was observed in the production time, indicating a slight adjustment in the machining process. Conversely, there was a slight decrease of 0.4% in the toolpath length in rapid motion, indicating a slight optimization in the efficiency of the machining operation. These findings suggest that the implementation of the proposed guidelines had a subtle impact on the end milling process, as no changes were impacted on the CAM programming of this operation after their implementation.

- **Profile Milling – Base Contour**

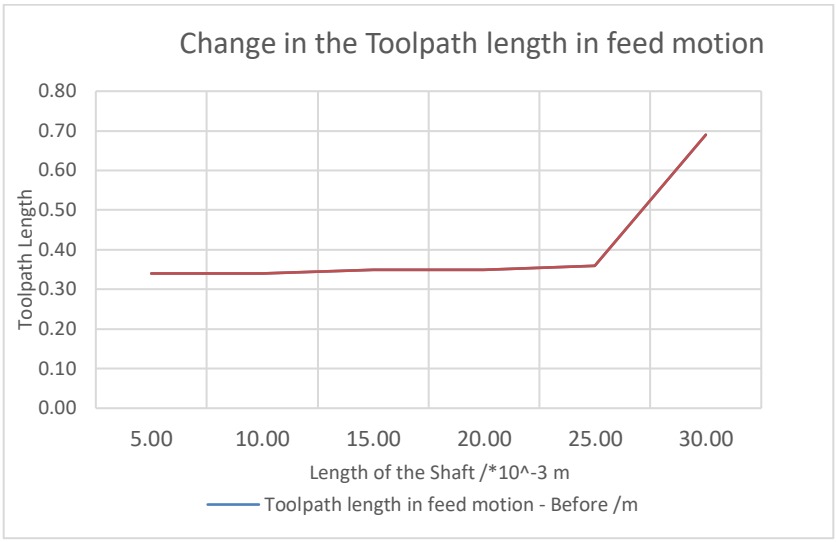


Figure 26: Change in the Toolpath in Feed Motion

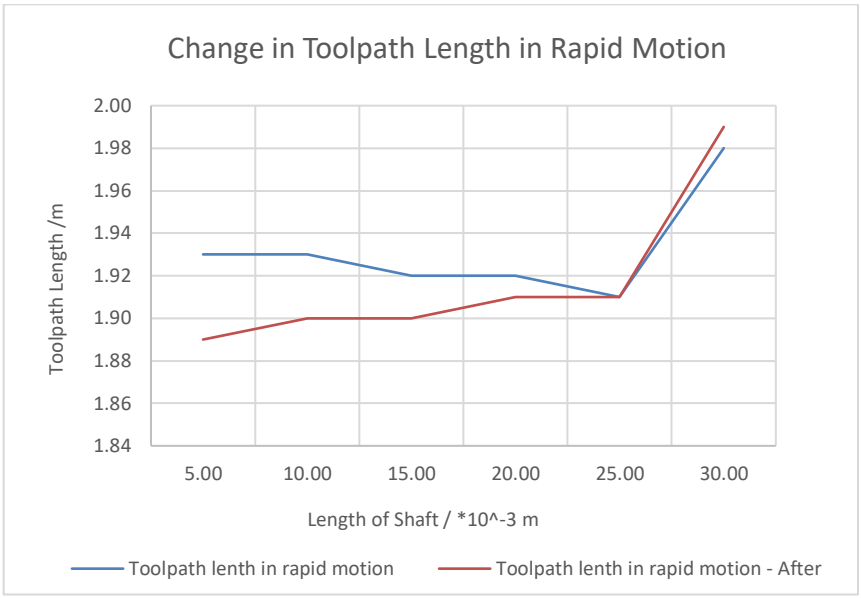


Figure 27: Change in the Toolpath Length in Rapid Motion

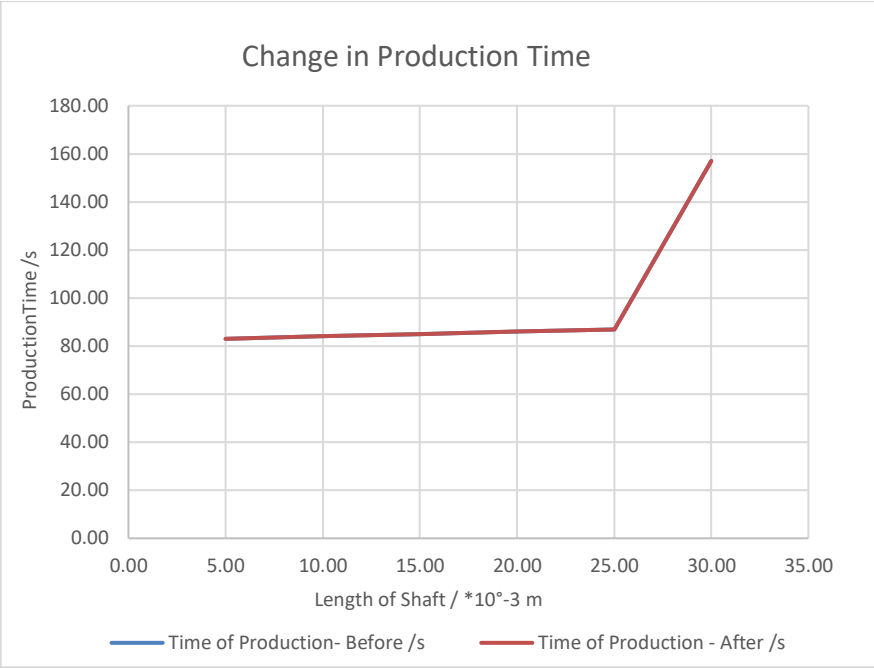


Figure 28: Change in Production Time

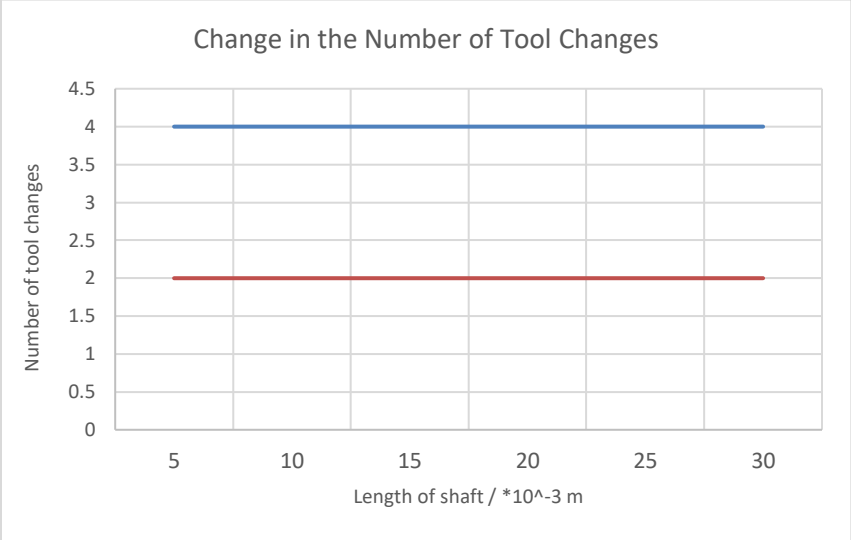


Figure 29: Change in the Number of Tool Changes

Similar observations were noted during the contour milling of the base of the flange. The implemented guidelines resulted in minimal alterations, with the primary change being a decrease in the toolpath length in rapid motion by 0.8%. This indicates a slight improvement in the efficiency of the contour milling operation, as the toolpath length was slightly reduced. Additionally, the number of tool changes reduced by 50%.

- **Profile Milling - Shaft Contour**

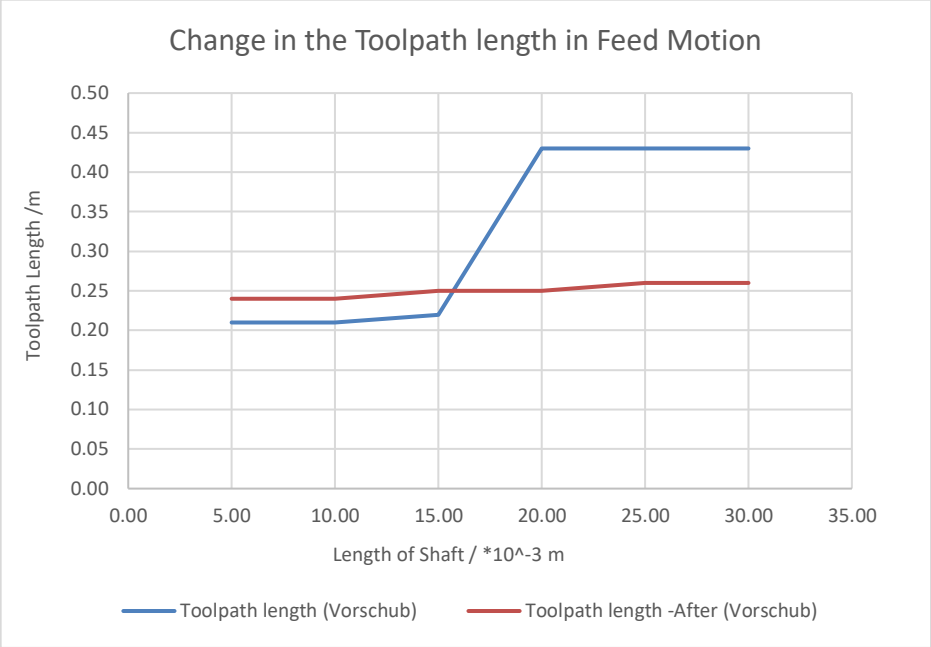


Figure 30: Change in the Toolpath Length

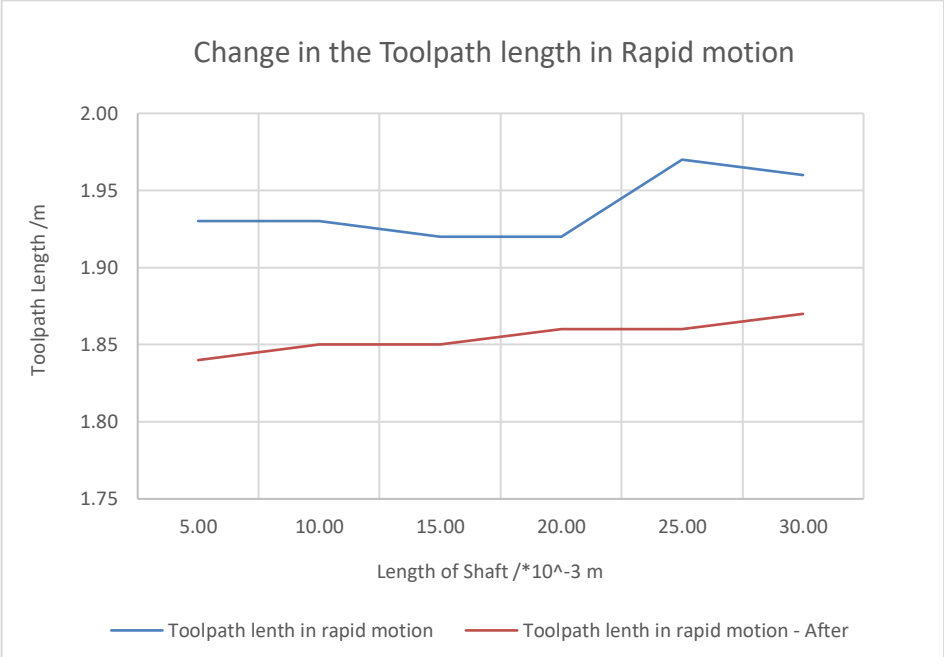


Figure 31: Change in the Toolpath Length in Rapid Motion

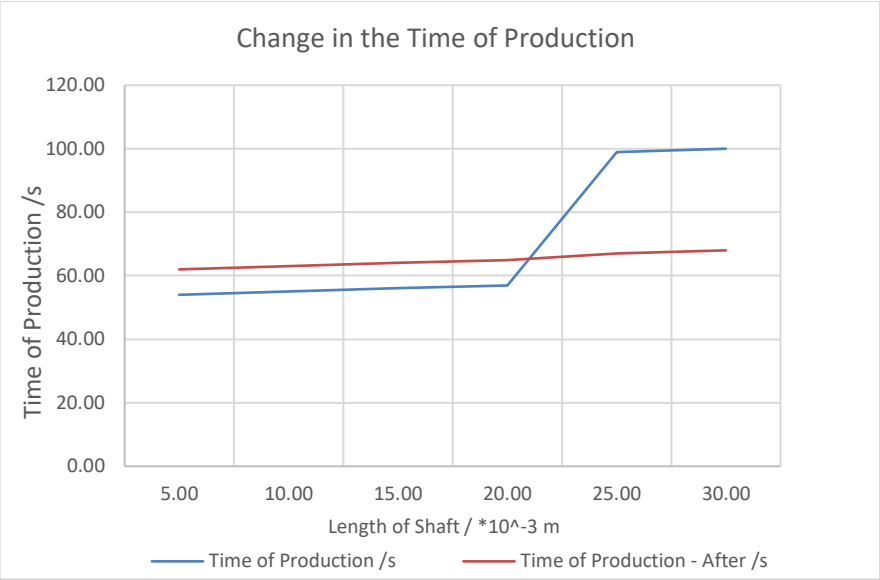


Figure 32: Change in Time of Production

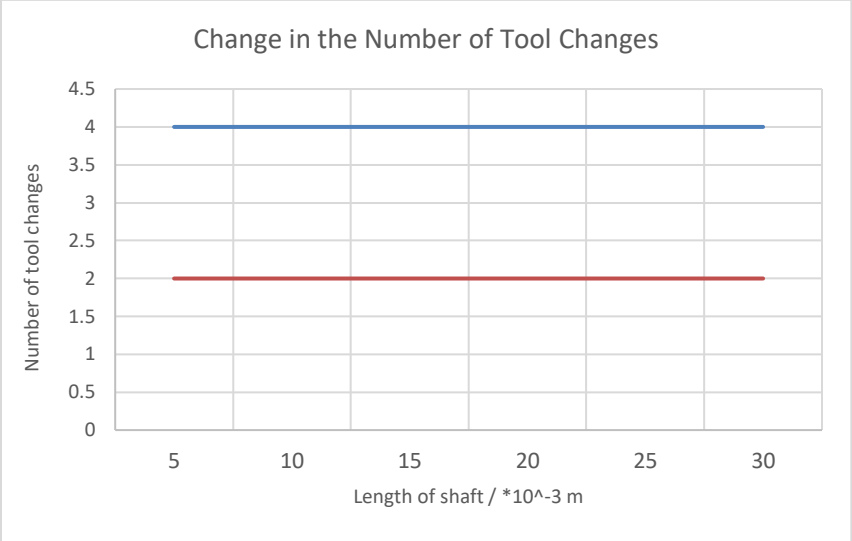


Figure 33: Change in the Number of Tool Changes

The implementation of the guidelines had a significant impact on this machining operation. The analysis revealed notable improvements in all key performance indicators. Specifically, the tool path in feed motion was reduced by 13.1%, indicating a more efficient and streamlined cutting path. Additionally, the tool path length in rapid motion experienced a reduction of 4.3%, indicating optimized tool movements. Also, the number of tool changes reduced by 50%. These improvements in tool path efficiency contributed to an overall reduction in production time by 1.1%.

The substantial reductions in tool path length and production time highlight the effectiveness of the guidelines in enhancing the machining operation for this specific process.

7 Procedure Proposal

7.1 Overview

The system's general framework is presented in Figure 34. The initial step requires the user to import or create a CAD model through software, such as SolidWorks. Once imported, the system evaluates the design and generates Design for Manufacturing (DFM) feedback for the user. If the feedback indicates no problematic features or if the identified problems are deemed not critical by the user, the user can proceed to the next step. However, if the feedback indicates significant issues, the user can modify the design based on the feedback provided and reanalyse the part. This iterative process continues until the user is satisfied with the design.

In the subsequent stage, the user is prompted to input the design parameters for the parts, such as geometry, dimensions, material type or tolerances. General tolerances can be specified for the entire part, and specific tolerances for features or configurations/variations of the standard part, requiring tighter or looser tolerances than the general tolerance. The system uses the user input values to assess the design and provide feedback. If the user has inputted extremely specific parameters such as tight tolerances or specific dimensions or geometry, the system will explain the implications of such parameters and confirm with the user if the input value is intended.

Once the user confirms satisfaction with the input design parameters, they can provide additional instructions for the manufacturers, including notes on desired surface finish, the number of required parts, and any special instructions. After the input of such details, the CAD model, along with input parameters and notes, is then submitted to the manufacturers, who utilize this information to produce the parts for the designers.

7.2 System Framework

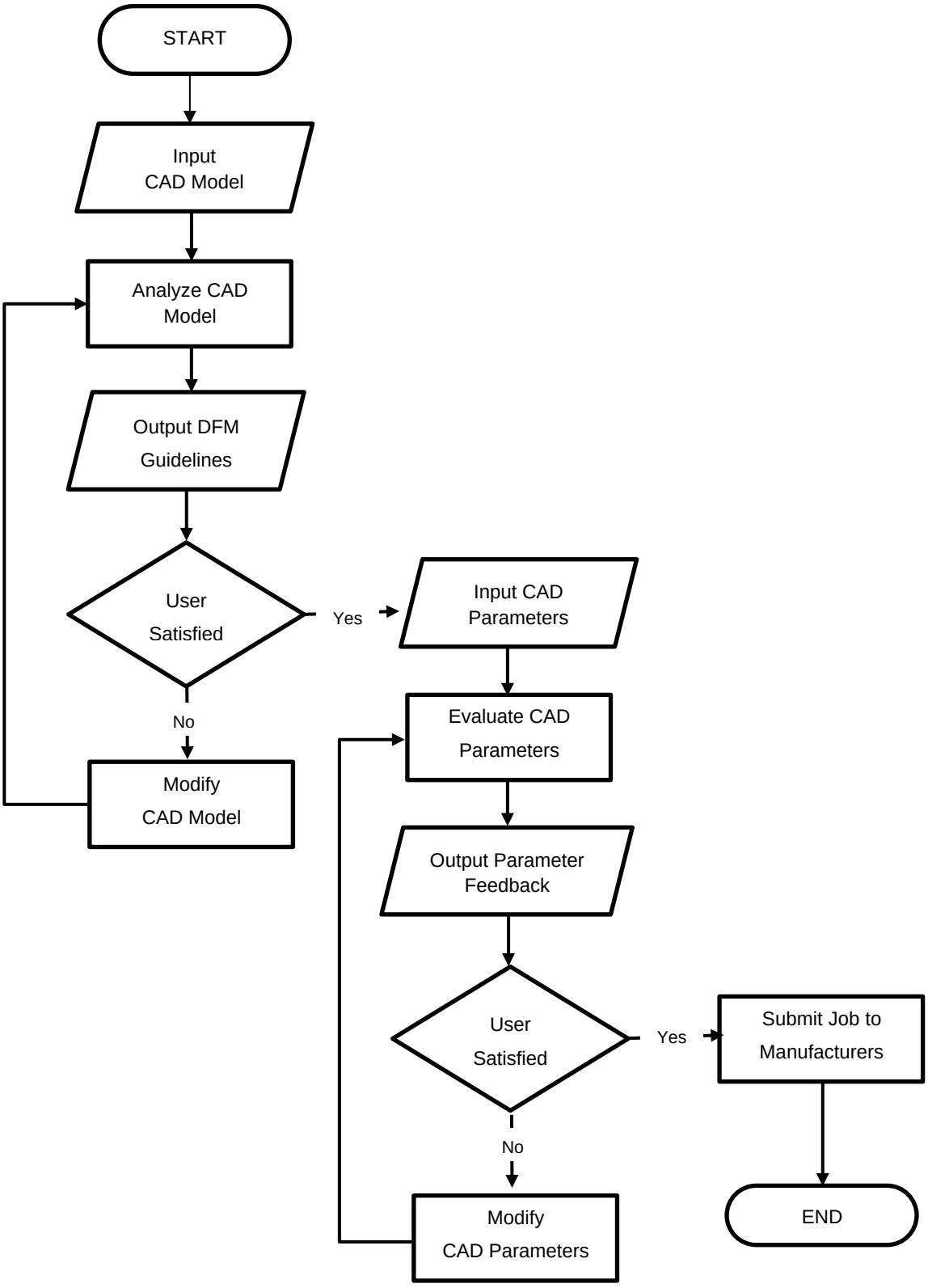


Figure 34: System Framework

7.3 Feedback Content and Acquisition Strategy

In order to provide designers with accurate and helpful feedback, the DFM software employs two different types of feedback. The first type of feedback is provided as an analysis of the CAD model using DFM rules. The second type of feedback is provided after the user has input design parameters, such as dimensions and geometry, and the software evaluates the design based on those parameters. The strategy used may include the visualization of problematic features by highlighting them and providing example suggestions showing how high manufacturability can be achieved.

For the first feedback, the software performs an analysis of the CAD model using DFM rules. The software can show on the CAD model which DFM rules have passed, and which have failed. For the rules that have failed, a pop-up dialogue explains the reasons for the failure when the designer puts the mouse on each rule. This allows the designer to quickly identify areas of the design that require further attention and modification.

Additionally, by clicking on a "Help" or "More Information" button, a DFM manual which includes example pictures showing high-manufacturability and low-manufacturability features for each rule could be portrayed. This provides designers with a clear understanding of what they need to do to ensure their designs meet DFM standards.

The second feedback is provided after the user has input design parameters such as dimension and geometry. The software evaluates the design based on these parameters and highlights any areas that need adjustment. When clicked upon, a dialog box pops up with information explaining the cause of the low manufacturability on the current parameter and providing suggestions on how to achieve high manufacturability.

It is important to note that the evaluation of the CAD model and the acquisition of the feedback information would be made by the analysis of the KPIs mentioned and examined in this work. These include tool path length, total production time, and number of tool changes. By analysing them, the software is able to provide designers with feedback that not only ensures their designs meet DFM standards, but also maximizes efficiency and cost-effectiveness during the manufacturing process.

8 Conclusion

8.1 Overview

The development of an automated procedure for providing design-accompanying manufacturing information would show promising results in enhancing machining production efficiency. The KPIs analysed, which included tool path length, both in feed and rapid motion, total production time, maximum diameter of usable cutting tool, and number of tool changes, have indicated a significant improvement in efficiency after implementation.

The procedure incorporated two feedback mechanisms: one for the analysis of DFM guidelines, and another for the analysis of inputted parameters. The DFM feedback provided designers with information about problematic features in the design, allowing for adjustments to be made to increase manufacturability. The inputted parameter feedback allowed for adjustments to be made to optimize the design for machining.

Overall, the developed procedure offers a promising solution for automating the provision of manufacturing information in the design process, leading to more efficient and effective manufacturing.

8.2 Limitations

A limitation of the study was encountered during the data collection process. The CAM software used, EXAPTsolid, did not support an automatic switch of the structure plan from one variation to another during CAM programming after designing separate variations of the part on SolidWorks. Consequently, each variation had to be manually adjusted and programmed to its respective parameters, resulting in a new variation, to permit an automatic refresh of the CAM structure plan on EXAPTsolid. This manual adjustment process was time-consuming and increased the risk of human error, potentially affecting the accuracy and reliability of the data collected. Therefore, this limitation should be considered when interpreting the results of the work.

Another limitation of this work is that some milling features were not automatically recognized by EXAPTsolid. As a result, these features had to be identified and manually inputted for each generated variation. This process was time-consuming and could have potentially led to errors or inconsistencies in the data collected. It is important for future work to address this limitation by improving the software's ability to automatically recognize all milling features, reducing the need for manual input, and increasing the efficiency of the data collection process.

8.3 Suggestions for Future Work

The first suggestion would be further research on the involvement and evaluation of parameters such as material properties and tolerances. The software, after further research on the procedure, should include these parameters, as they are important factors in CAD, CAM, and the world of CNC machining.

The second suggestion would be the integration of the procedure with CAD software, such as SolidWorks. This would make it easily accessible to designers and manufacturers alike, simplifying the communication process and improving the overall efficiency of the manufacturing process. The system's ability to analyse design and evaluate manufacturability, coupled with its automated feedback mechanisms, would provide an efficient and effective means of enhancing the design process and reducing errors.

Future research could focus on expanding the scope of KPIs analysed to include additional factors that may affect manufacturing efficiency, as well as refining the feedback mechanisms to further enhance the system's ability to provide accurate and useful feedback to designers.

Nonetheless, the automated procedure developed in this study is a significant step forward in enhancing the manufacturing process and reducing design errors, ultimately resulting in improved efficiency and reduced costs.

9 Bibliography

(1) Forschungsvereinigung Programmiersprachen für Fertigungseinrichtungen: Aufgaben
https://www.forschungsvereinigung-programmiersprachen.de/de/

(2) EXAPT Systemtechnik GmbH
https://www.exapt.de/de/unternehmen#wir_ueber_uns

(3) Wikipedia, Computer Aided Design:
https://en.wikipedia.org/wiki/Computer-aided_design

(4) How does CAD work? What is Computer Aided Design?
https://www.brighthubengineering.com/cad-autocad-reviews-tips/67118-how-does-cad-work/

(5) Wikipedia, Computer Aided Manufacturing:
https://en.wikipedia.org/wiki/Computer-aided_manufacturing

(6) What is Computer Aided Manufacturing (CAM)?
https://fractory.com/what-is-computer-aided-manufacturing-cam/

(7) Towards Automated Guidance for Helping Novices Design for Sustainable Additive Manufacturing and CNC Machining, 2019, Ruoyu Song

(8) Design for Manufacturing – Guidelines, Information taken from Computer-Aided Manufacturing, Second Edition, Tien-Chien Chang, Richard A Wysk, and Hsu-Pin Wang. Pages 596 to 598. Prentice Hall 1998
https://www.unm.edu/~bgreen/ME101/dfm.pdf

(9) The Ultimate CNC Machining Design for Manufacturing Checklist, Fast Radius Incorporated, 2021, Pages

(10) Automated Manufacturability Analysis: A Survey, S. K. Gupta, W. C. Regli, Diganta Das, Dana S. Nau

(11) Manufacturability Analysis Systems: Review Issues and Future Trends, Syaimak Abdul Shukor, D. Axinte, Pages 2 10 15.

(12) What Is CNC Machining? Working Principles, Capabilities & More
https://fractory.com/what-is-cnc-machining/#G-Code_vs_M-Code

(13) Identifying KPIs for the design stage based on the main design subprocesses, Haponova Tatsiana

List of Figures

FIGURE 1: COMMUNICATION BETWEEN DESIGNERS AND MANUFACTURERS (SONG, 2019).....3.9
FIGURE 2: INITIAL DESIGN OF THE FLANGE.....4.13
FIGURE 3: SIMULATION OF THE RAW PART.....4.15
FIGURE 4: EXAPTSOLID STRUCTURAL PLAN FOR FLANGE CONFIGURATION.....4.17
FIGURE 5: DATA REPRESENTATION ON EXAPTPLUS 5.19
FIGURE 6: DATA REPRESENTATION ON EXAPTPLUS EMPHASIZING ON THE NUMBER OF TOOL CHANGES 5.19
FIGURE 7: GRAPHICAL REPRESENTATION: VARIATION IN THE NUMBER OF HOLES 5.21
FIGURE 8: GRAPHICAL REPRESENTATION - VARIATION IN THE NUMBER OF POCKETS.....5.23
FIGURE 9: GRAPHICAL REPRESENTATION - END MILLING OF THE TOP OF THE FLANGE 5.25
FIGURE 10: GRAPHICAL REPRESENTATION - BASE CONTOUR MILLING.....5.25
FIGURE 11: GRAPHICAL REPRESENTATION - SHAFT CONTOUR MILLING.....5.26
FIGURE 12: DESIGN AFTER GUIDELINE IMPLEMENTATION 6.28
FIGURE 13: CHANGE IN THE TOOLPATH LENGTH IN FEED MOTION.....6.31
FIGURE 14: CHANGE IN PRODUCTION TIME.....6.31
FIGURE 15: CHANGE IN THE TOOLPATH LENGTH IN RAPID MOTION.....6.32
FIGURE 16: CHANGE IN THE NUMBER OF TOOL CHANGES 6.32
FIGURE 17: CHANGE IN THE TOOLPATH LENGTH IN FEED MOTION.....6.34
FIGURE 18: CHANGE IN PRODUCTION TIME.....6.34
FIGURE 19: CHANGE IN THE TOOLPATH LENGTH IN RAPID MOTION.....6.35
FIGURE 20: CHANGE IN THE MAXIMUM RADIUS OF CUTTING TOOL.....6.35
FIGURE 21: CHANGE IN THE NUMBER OF TOOL CHANGES 6.36
FIGURE 22: CHANGE IN THE TOOLPATH LENGTH IN FEED MOTION.....6.38
FIGURE 23: CHANGE IN THE TOOLPATH LENGTH IN RAPID MOTION.....6.39
FIGURE 24: CHANGE IN THE PRODUCTION TIME 6.39
FIGURE 25: CHANGE IN THE NUMBER OF TOOL CHANGES 6.40
FIGURE 26: CHANGE IN THE TOOLPATH IN FEED MOTION 6.41
FIGURE 27: CHANGE IN THE TOOLPATH LENGTH IN RAPID MOTION.....6.41
FIGURE 28: CHANGE IN PRODUCTION TIME.....6.42
FIGURE 29: CHANGE IN THE NUMBER OF TOOL CHANGES 6.42
FIGURE 30: CHANGE IN THE TOOLPATH LENGTH 6.43
FIGURE 31: CHANGE IN THE TOOLPATH LENGTH IN RAPID MOTION.....6.43
FIGURE 32: CHANGE IN TIME OF PRODUCTION 6.44
FIGURE 33: CHANGE IN THE NUMBER OF TOOL CHANGES 6.44
FIGURE 34: SYSTEM FRAMEWORK.....7.46

List of Tables

TABLE 1: TOOLS AND CORRESPONDING IDS IN EXAPT DATABASE 4.16

TABLE 2: DATA COLLECTION - VARIATION IN NUMBER OF HOLES 5.20

TABLE 3: VARIATION IN THE NUMBER OF POCKETS - DATA COLLECTION 5.22

TABLE 4: DATA COLLECTION - END MILLING FOR THE TOP OF THE FLANGE.....5.24

TABLE 5: DATA COLLECTION - PROFILE MILLING FOR THE BASE OF THE FLANGE.....5.24

TABLE 6: DATA COLLECTION - PROFILE MILLING FOR THE SHAFT OF THE FLANGE.....5.24

TABLE 7: TOOLS AND CORRESPONDING IDS IN EXAPT DATABASE 6.29

TABLE 8: DATA COLLECTION - VARIATION IN THE NUMBER OF HOLES AFTER IMPLEMENTATION.....6.30

TABLE 9: DATA COLLECTION - VARIATION IN THE NUMBER OF POCKETS AFTER IMPLEMENTATION 6.33

TABLE 10: DATA COLLECTION – END MILLING FOR THE TOP OF THE FLANGE.....6.37

TABLE 11: DATA COLLECTION – PROFILE MILLING FOR THE BASE OF THE FLANGE.....6.37

TABLE 12: DATA COLLECTION – PROFILE MILLING FOR THE SHAFT OF THE FLANGE.....6.38